

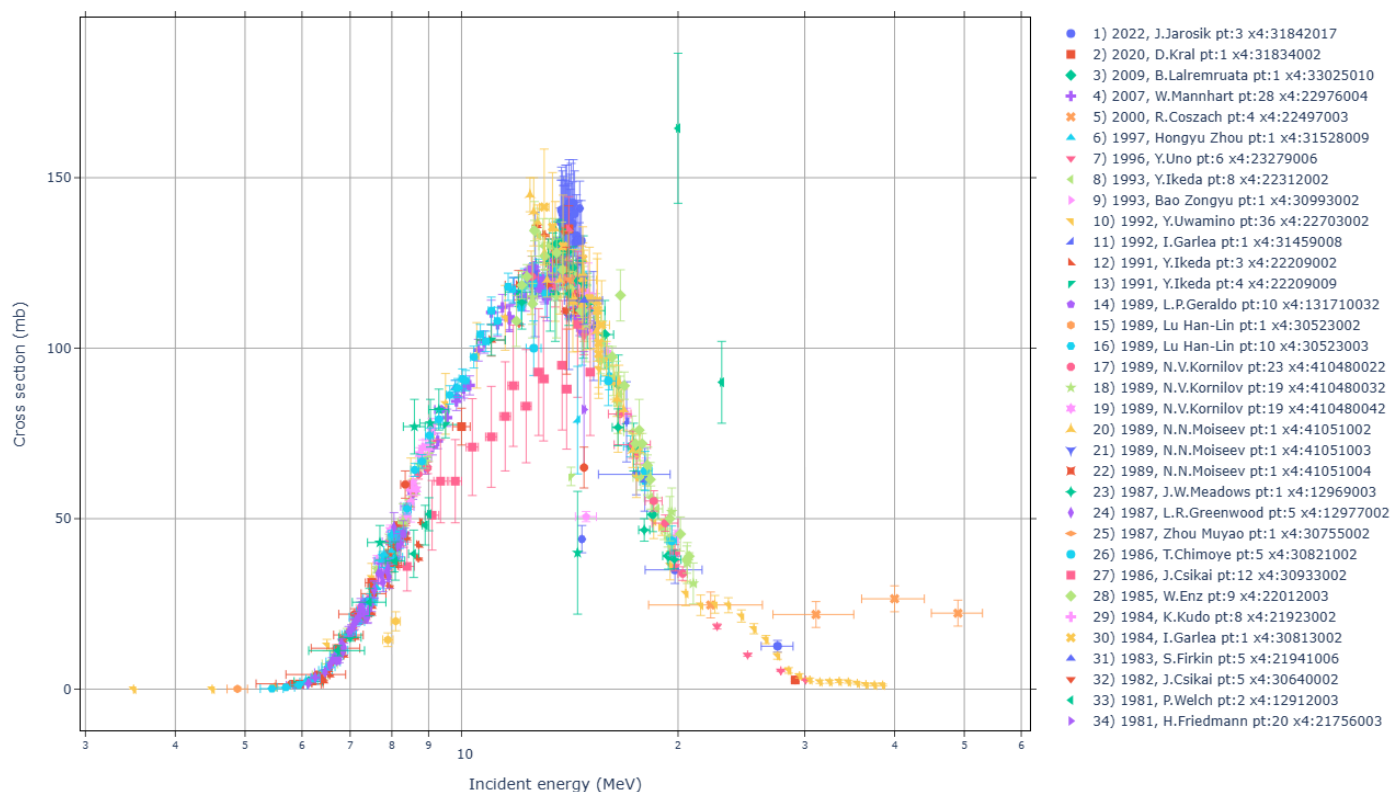
X4Pro. Plotting via Plotly.

part1-1-sig/ Retrieve and plot EXFOR cross sections (modular Python-program)

1) part1-1-sig/out00/Al27na-1.html.png

EXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

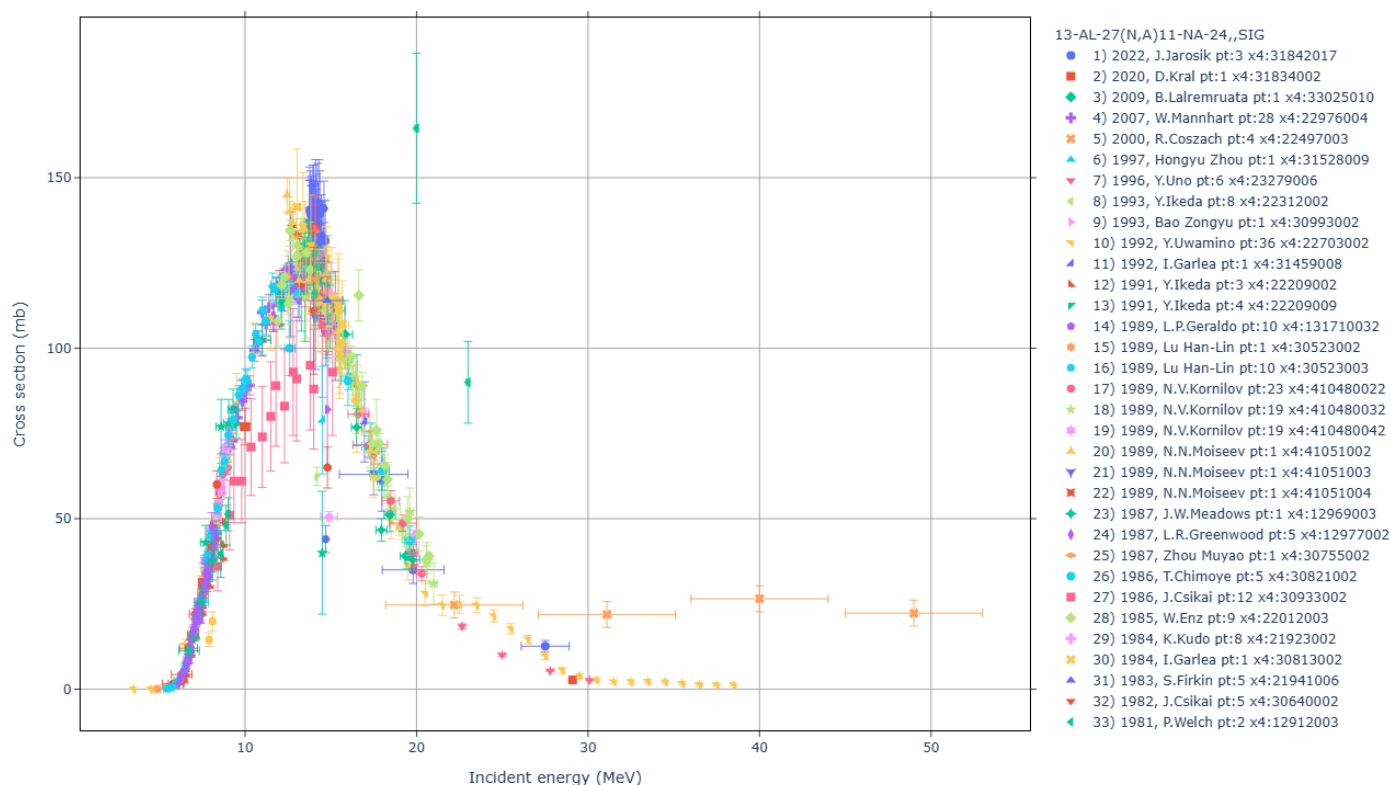
X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:03:38



2) part1-1-sig/out00/Al27na-2.html.png

EXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:03:42

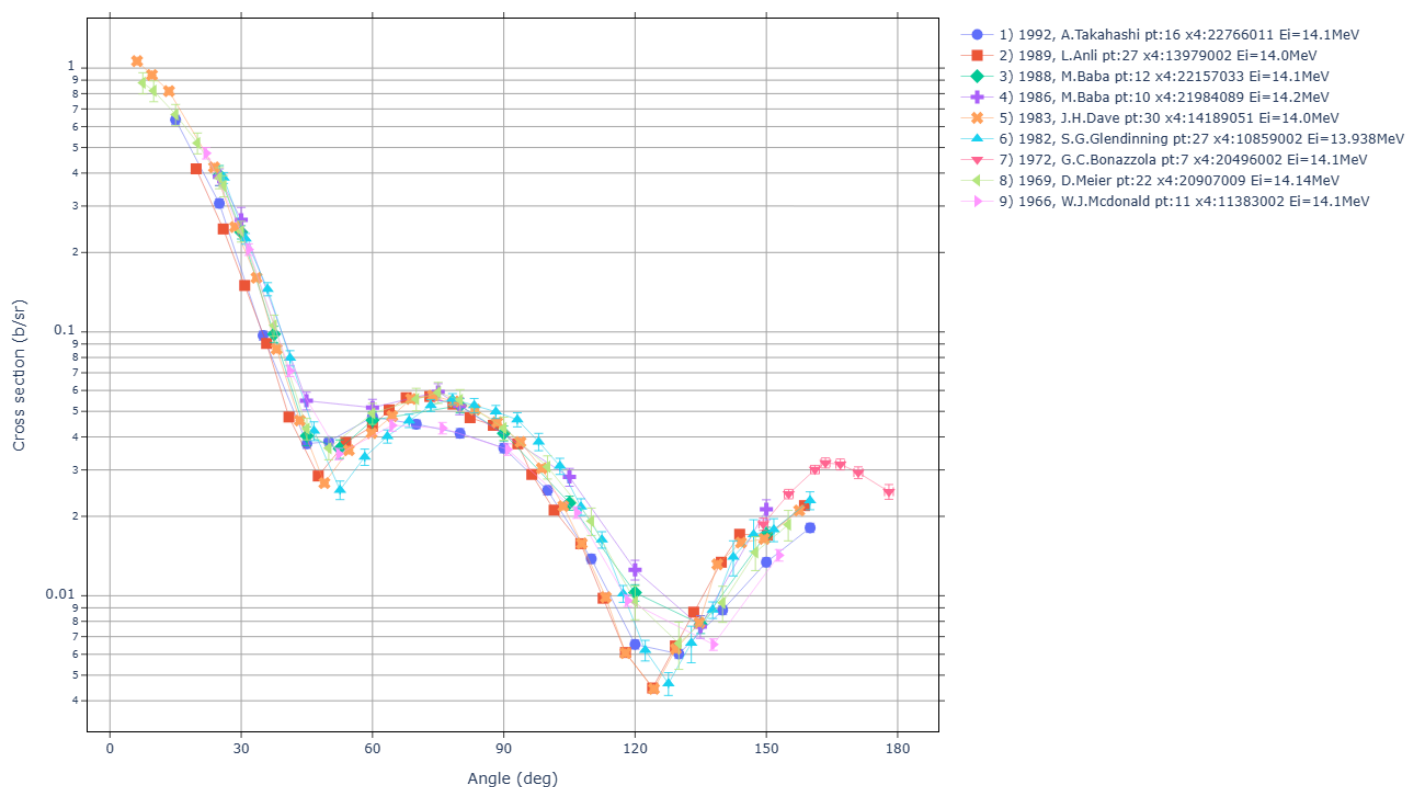


part1-2-da/ EXFOR angular distributions

3) part1-2-da/out00/da0an.html.png

Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-12-20 //running:2026-08-21 12:03:45

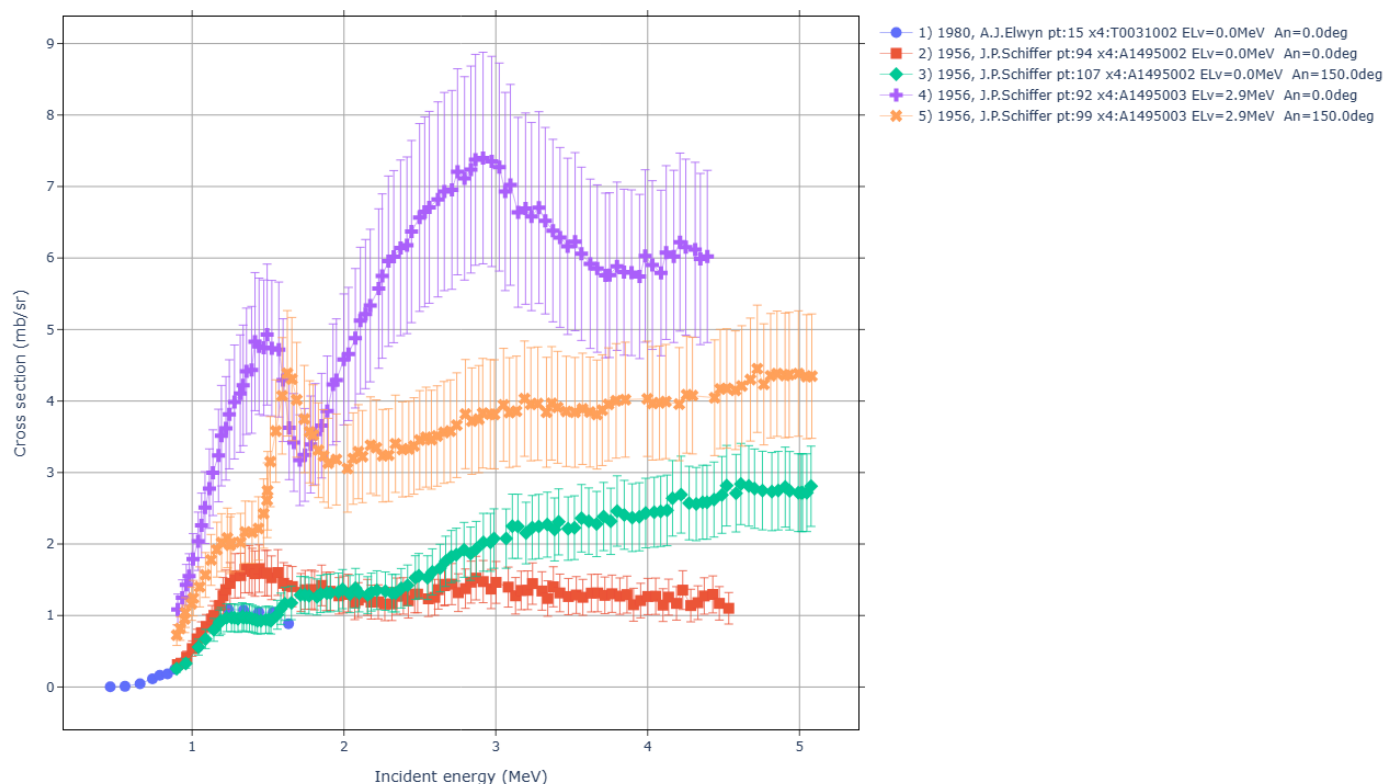


part1-3-dap/ EXFOR partial angular distributions

4) part1-3-dap/out00/dap0ei.html.png

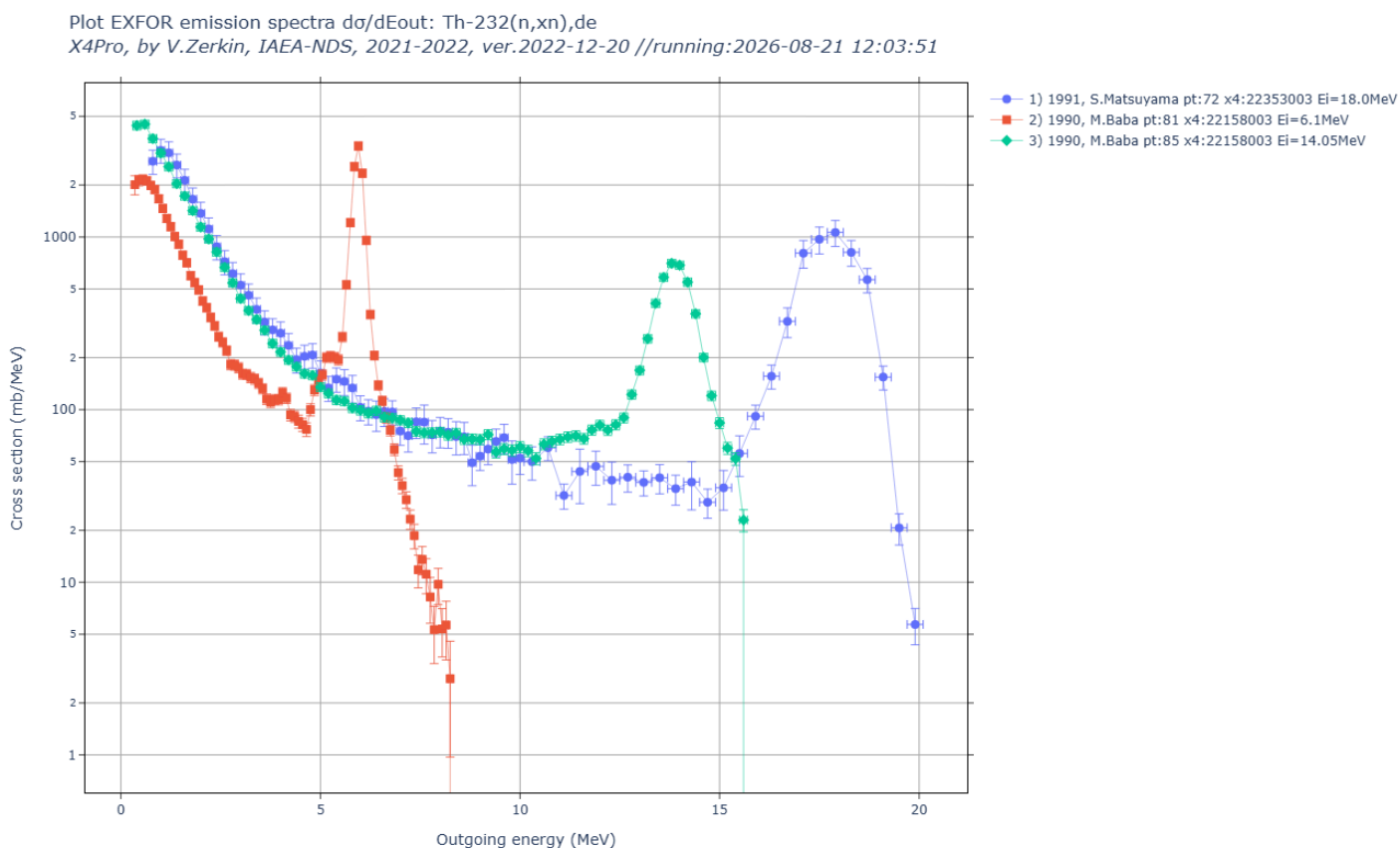
Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: Li-6(he3,p)par,da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:03:48



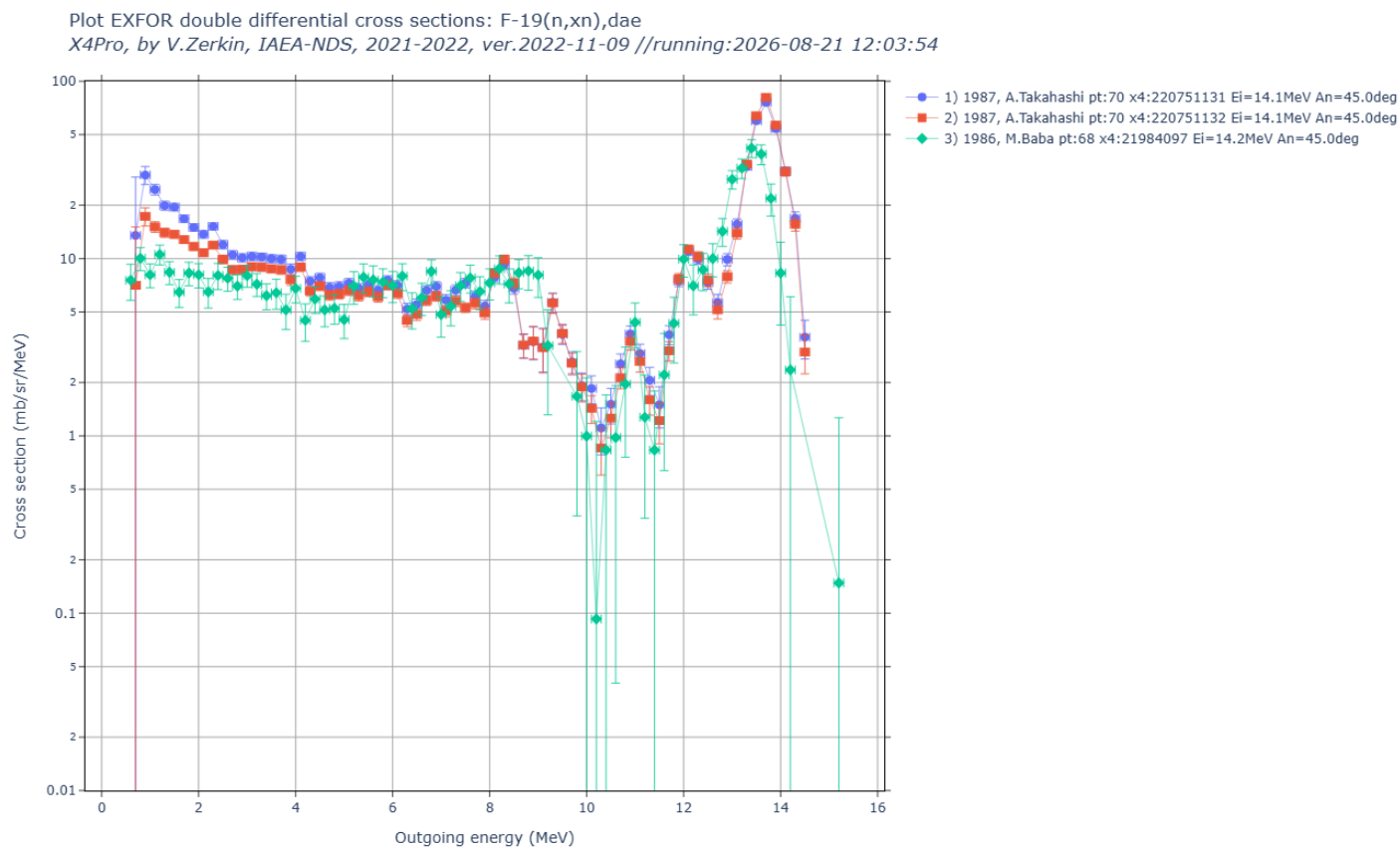
part1-4-de/ EXFOR energy spectra of outgoing particles

5) part1-4-de/out00/de0eo.html.png



part1-5-dae/ EXFOR double differential cross sections

6) part1-5-dae/out00/dae0eo.html.png

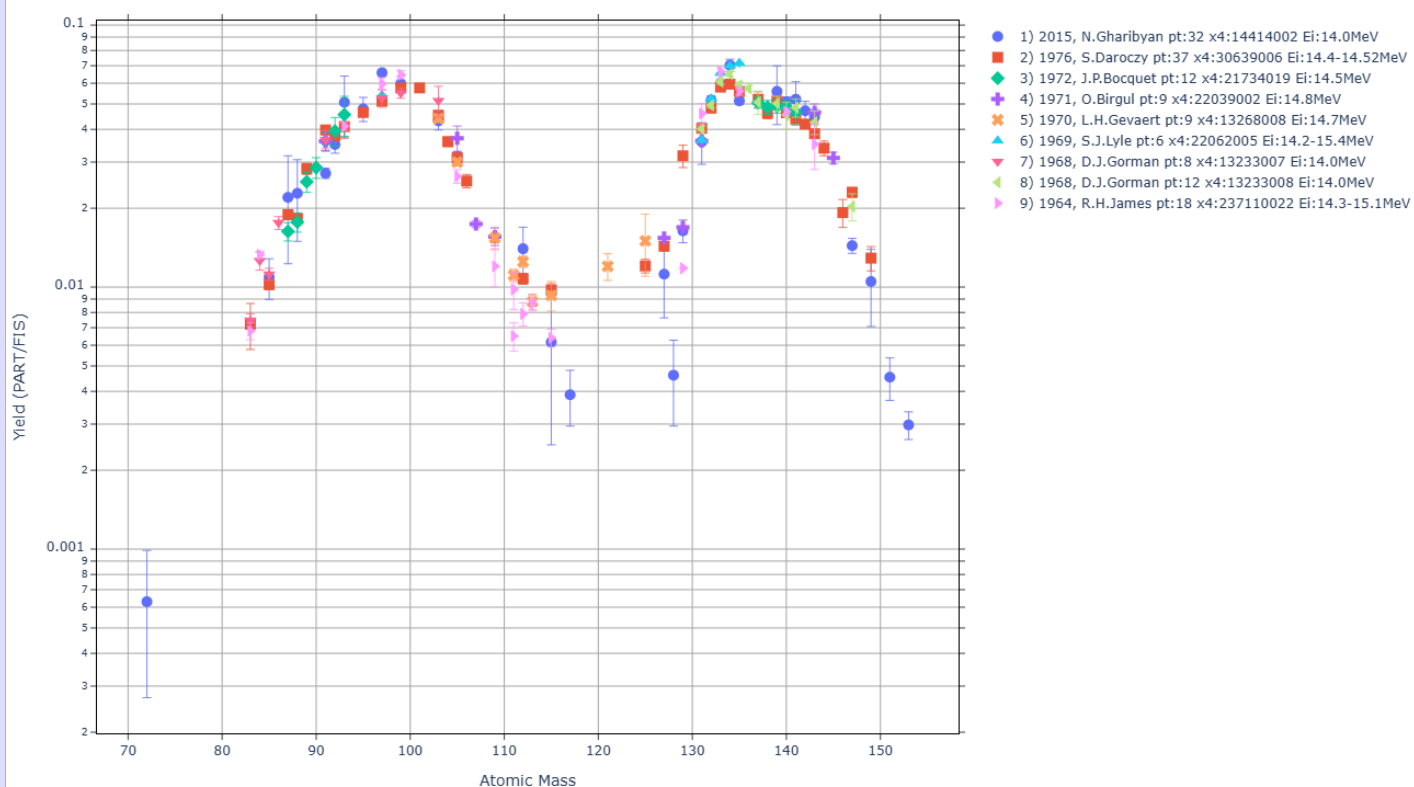


part1-6-fy/ EXFOR fission yield

7) part1-6-fy/out00/U238nf-fy.html.png

EXFOR fission yield FY(A,Ei): 92-U-238(N,F)MASS,CHN,FY

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:03:57

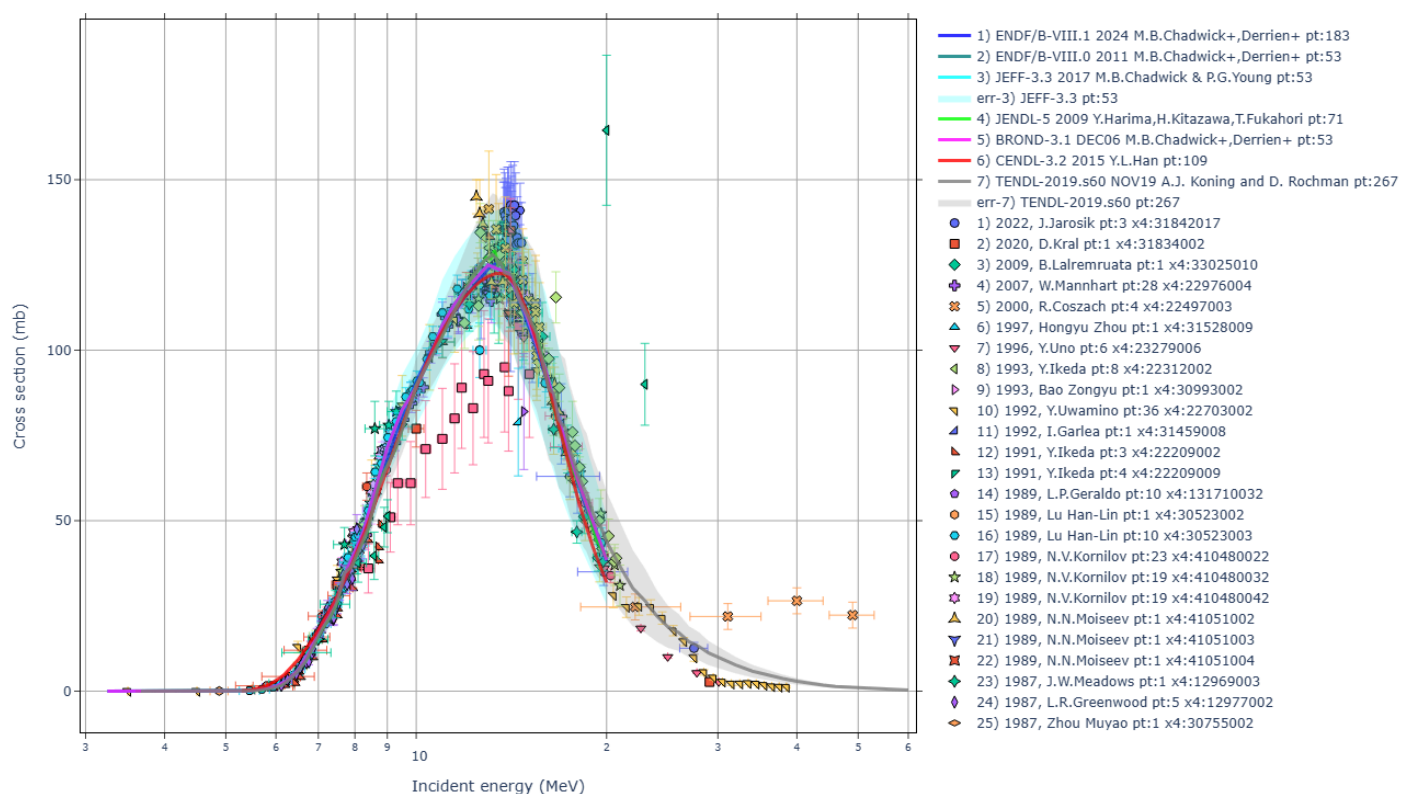


part2-1-sig1/ Retrieve and plot EXFOR/ENDF cross sections

8) part2-1-sig1/out00/Al27na.html.png

EXFOR/ENDF cross sections $\sigma(E)$: Al-27(n,a),sig EXFOR-datasets:87

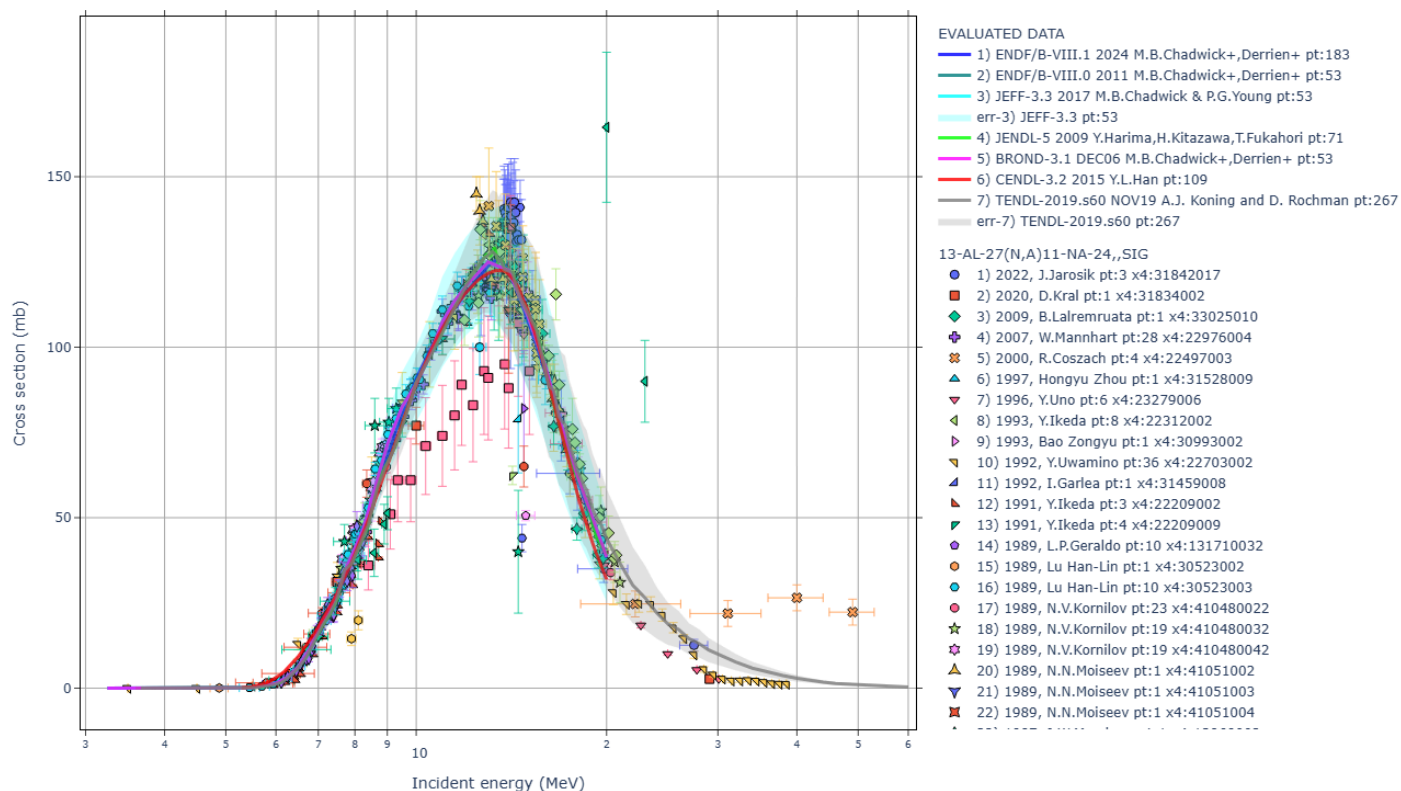
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-08-21 12:04:30



9) part2-1-sig1/out00/A127na2.html.png

EXFOR/ENDF cross sections $\sigma(E)$: Al-27(n,a),sig EXFOR-datasets:93

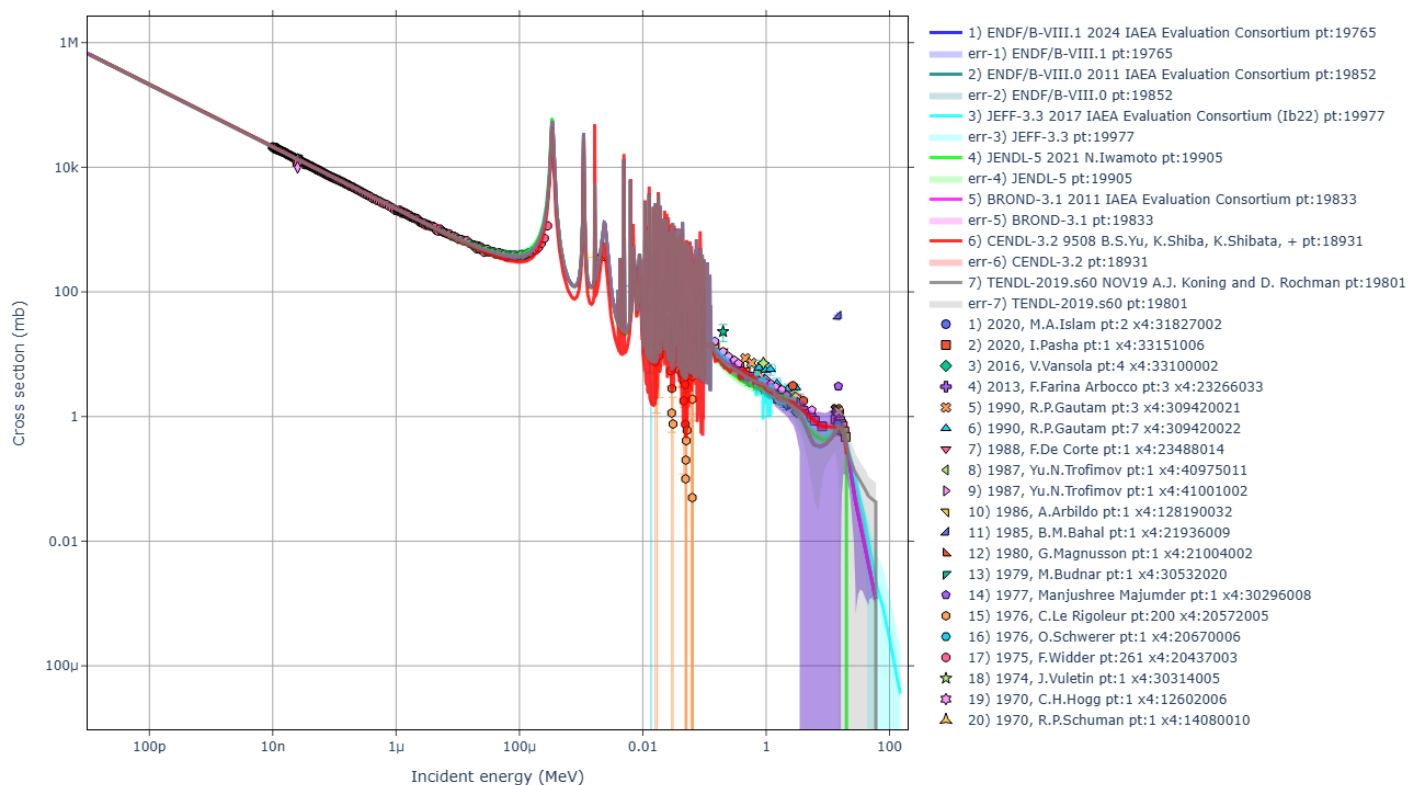
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-08-21 12:04:56



10) part2-1-sig1/out00/mn55ng.html.png

EXFOR/ENDF cross sections $\sigma(E)$: mn-55(n,g),sig EXFOR-datasets:49

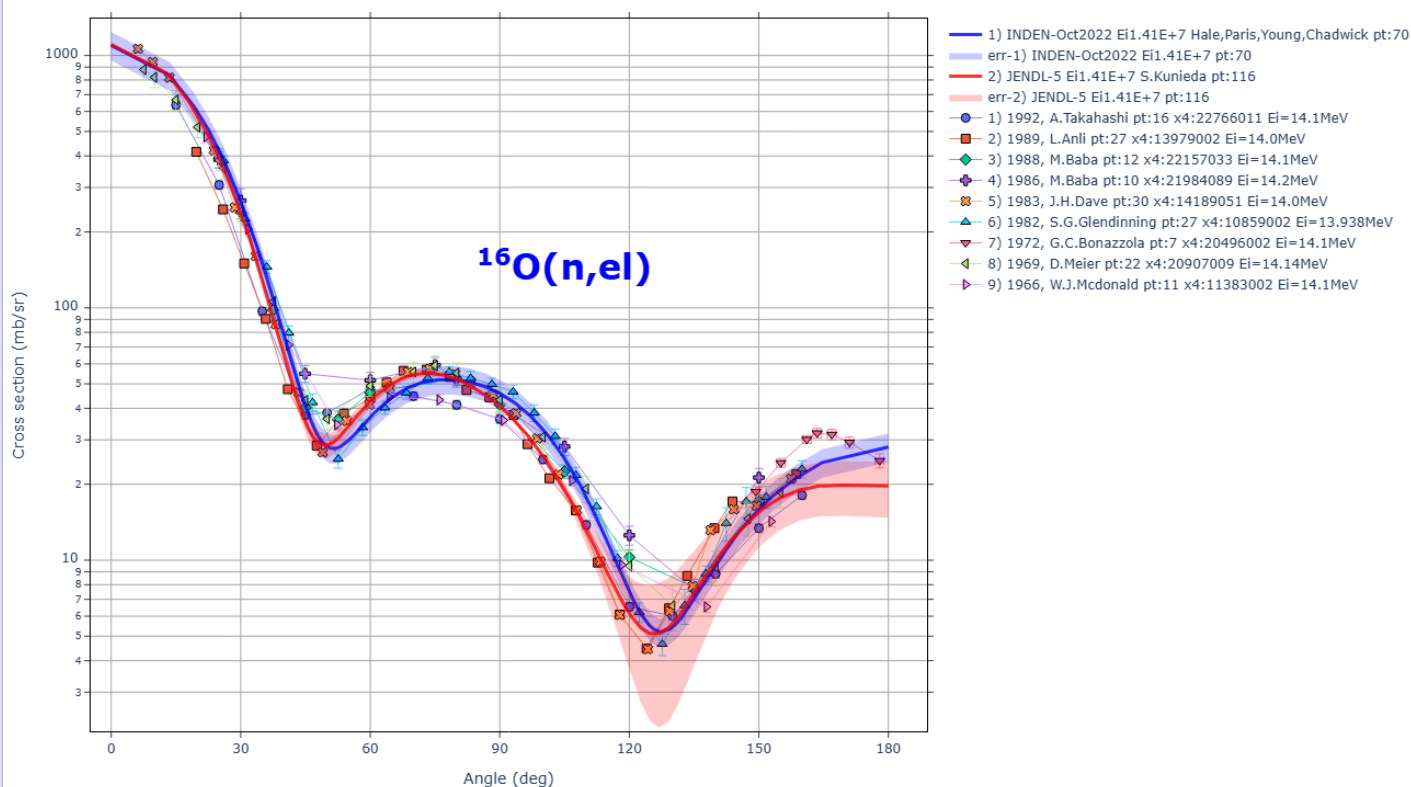
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-08-21 12:04:59



part2-2-dalán/ EXFOR/ENDF angular distributions (E_i :fixed)

11) part2-2-dalán/out00/dalán.html.png

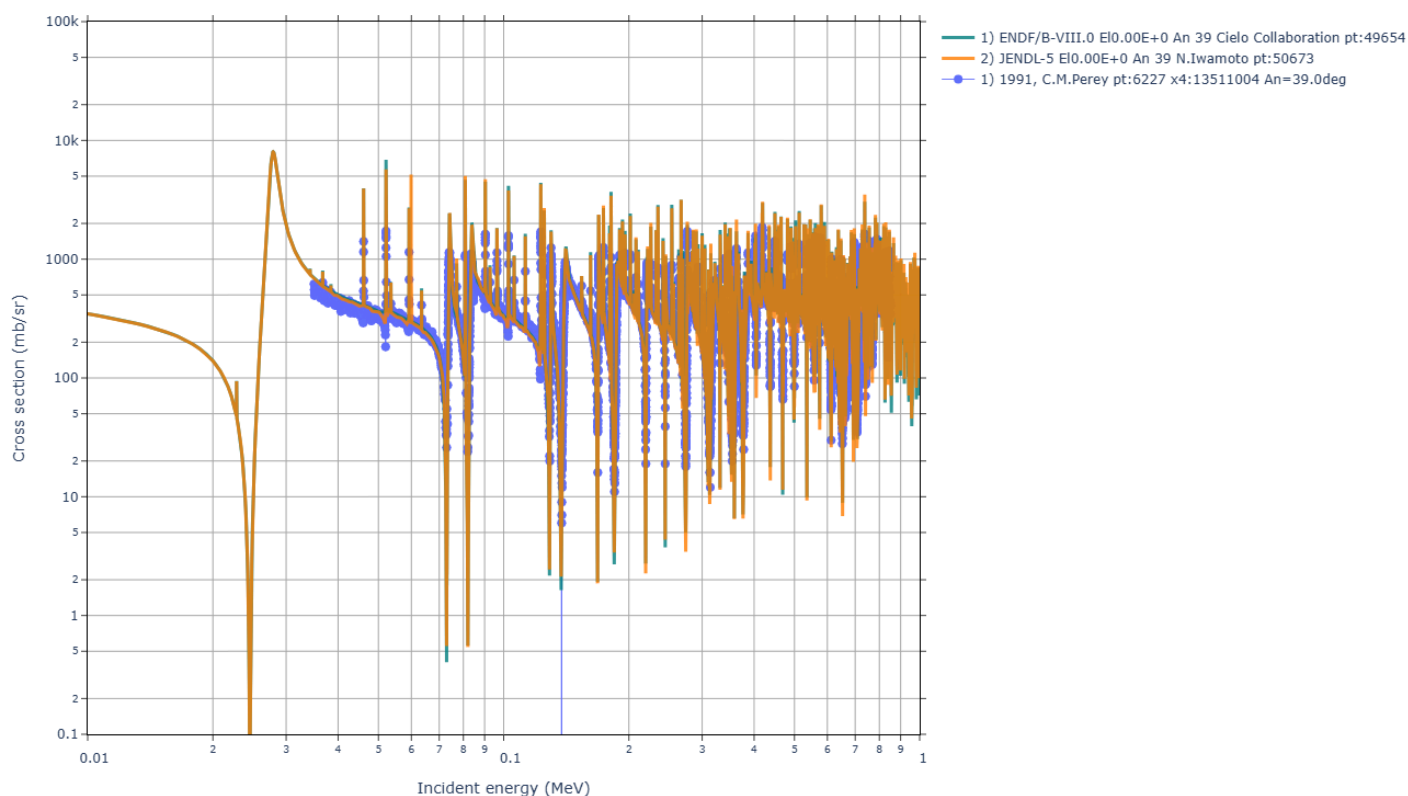
EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da EXFOR-datasets:9
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-08-21 12:05:31



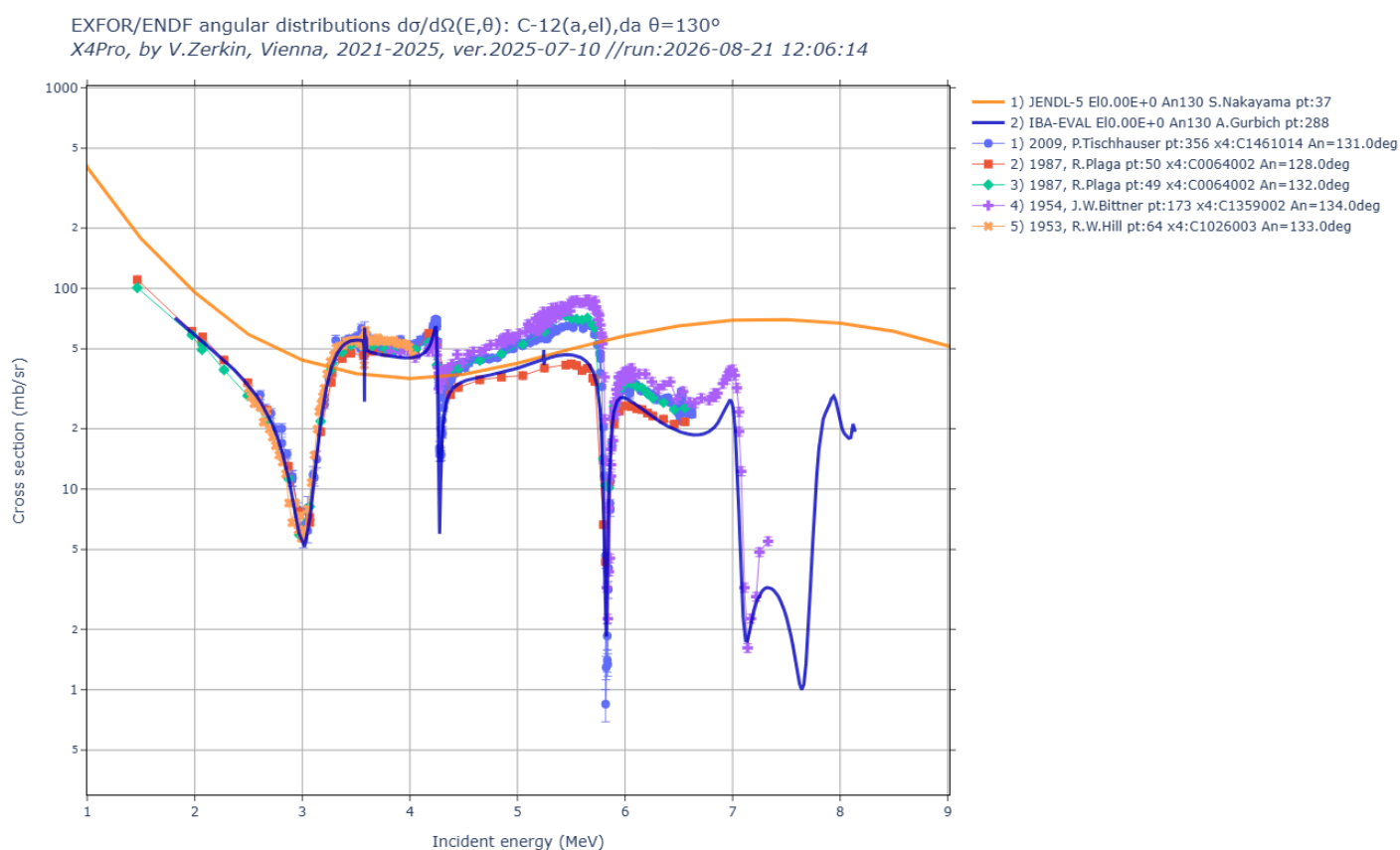
part2-3-daléi/ EXFOR/ENDF angular distributions (angle:fixed)

12) part2-3-daléi/out00/daléi-ex2.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: Fe-56(n,el),da $\theta=39^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-08-21 12:06:26

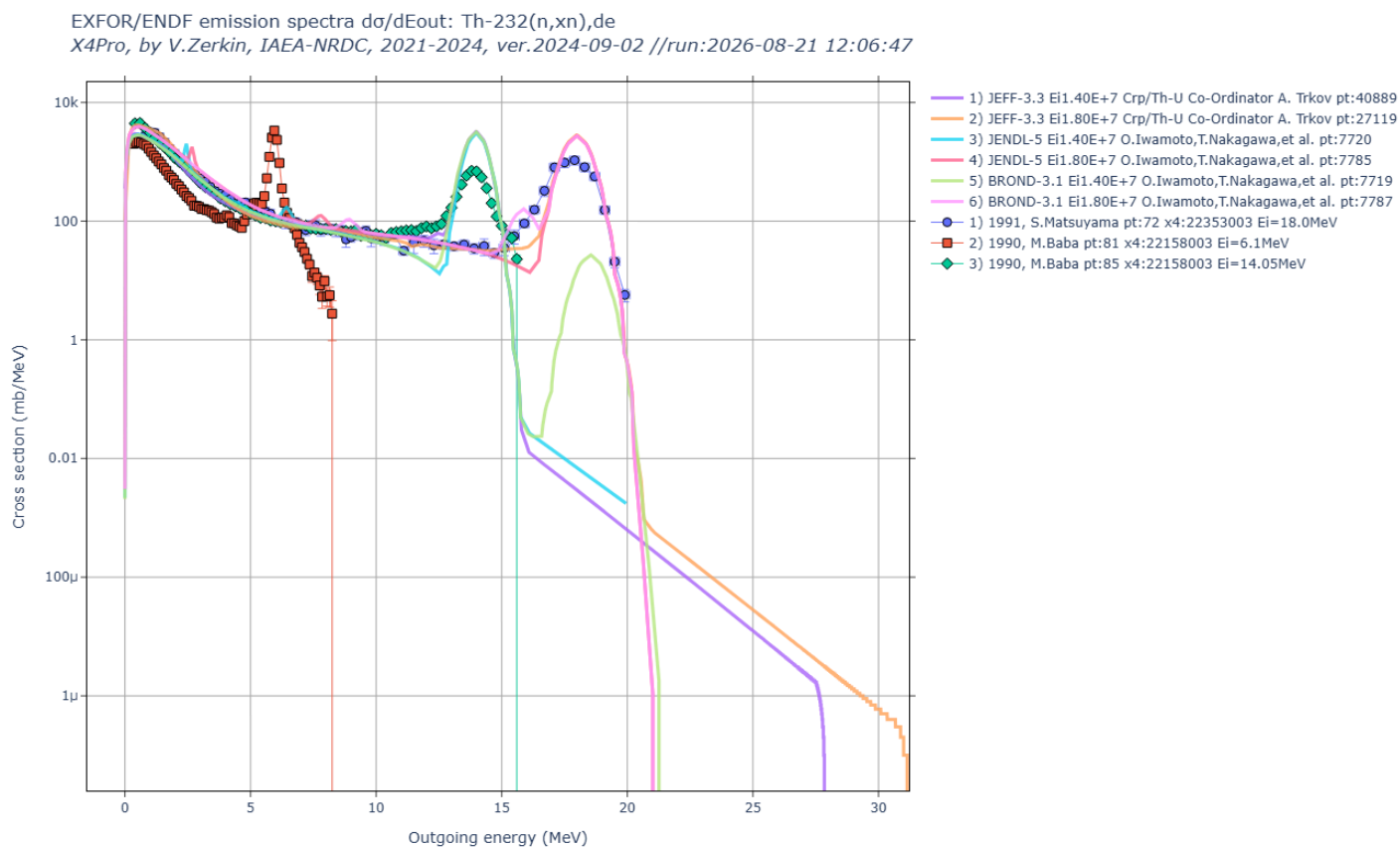


13) part2-3-da1ei/out00/da1ei.html.png



part2-4-de1eo/ EXFOR/ENDF energy spectra of outgoing particles

14) part2-4-de1eo/out00/de1eo.html.png

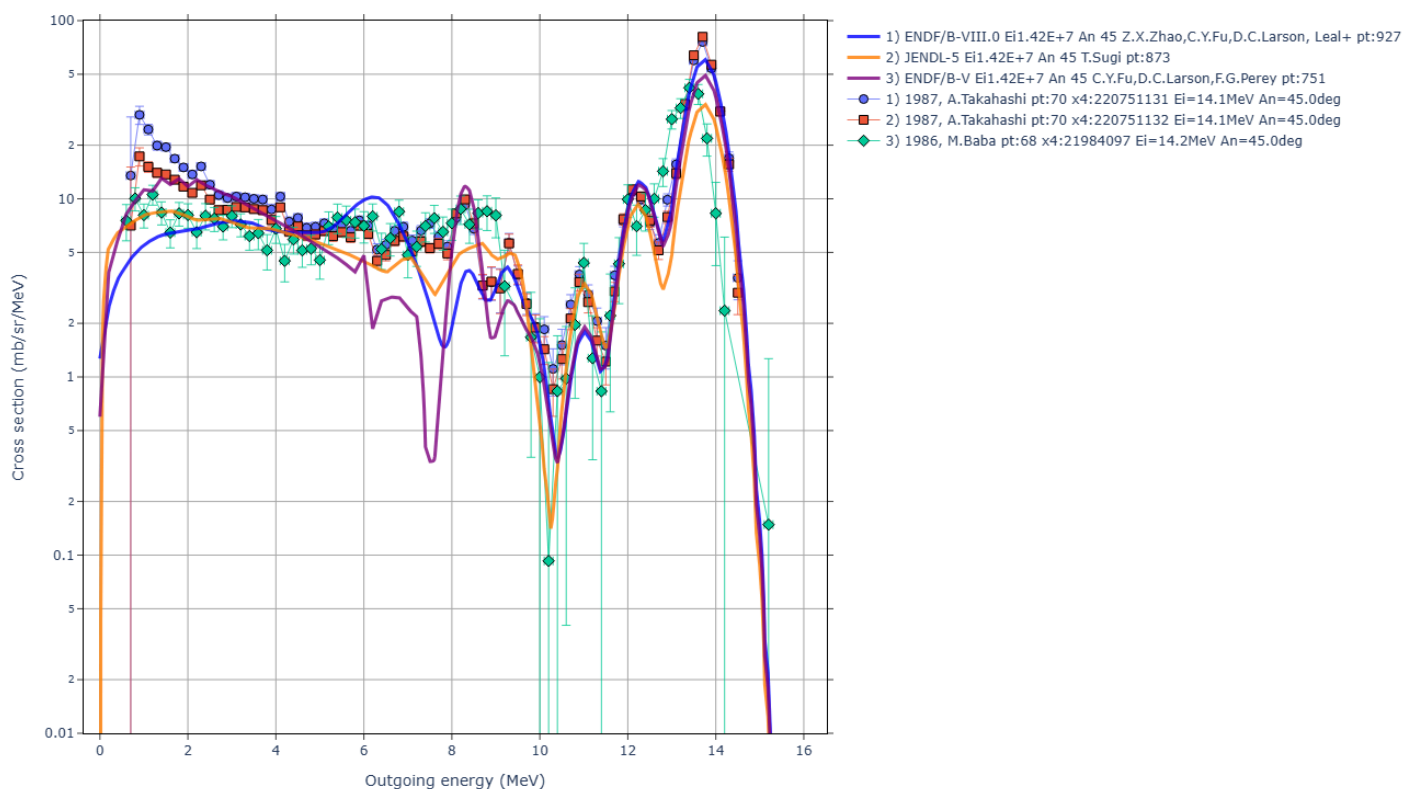


part2-5-dae1eo/ EXFOR/ENDF double differential cross sections

15) part2-5-dae1eo/out00/dae1eo.html.png

EXFOR/ENDF double differential cross sections: F-19(n,xn),dae

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2024, ver.2024-09-02 //run:2026-08-21 12:07:56

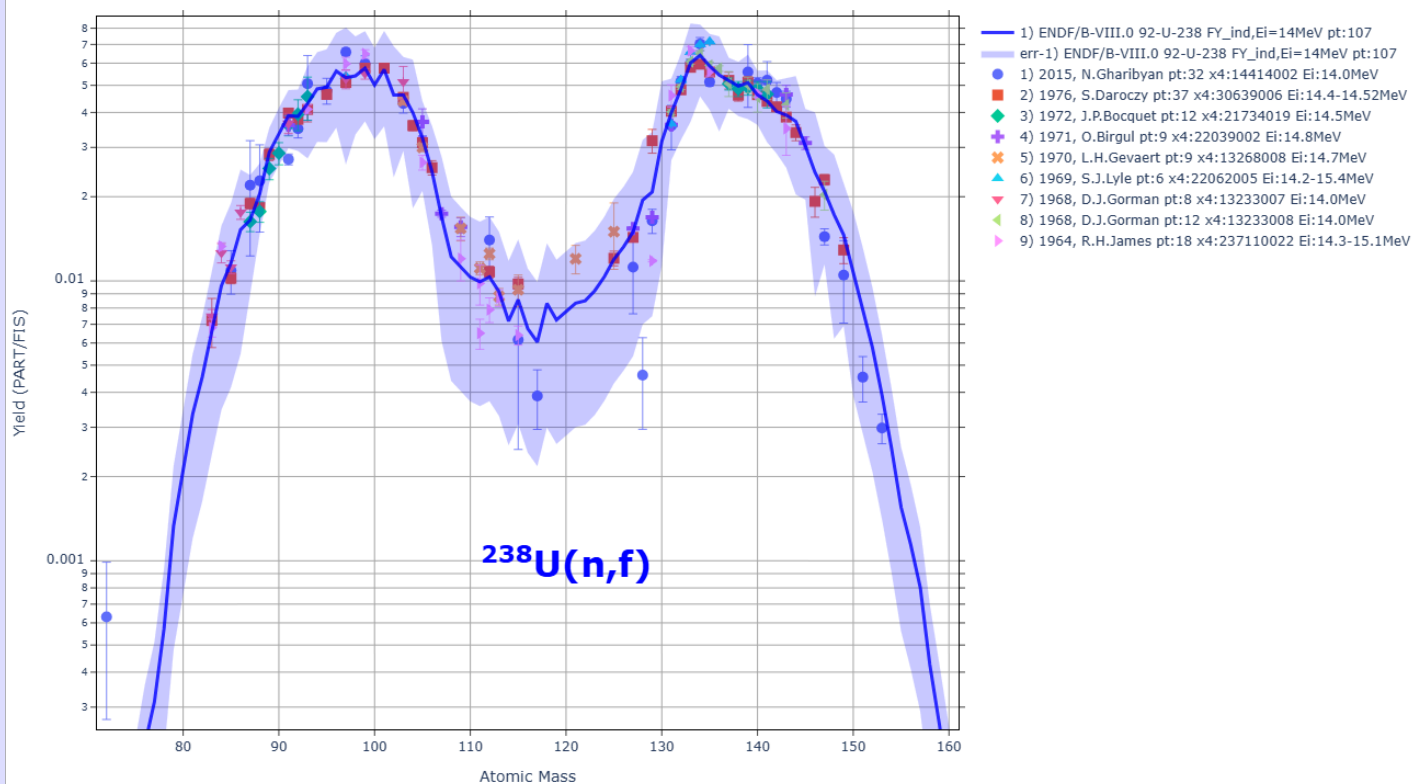


part2-6-fy/ EXFOR/ENDF fission yeild (mass distribution)

16) part2-6-fy/out00/U238nf-fy.html.png

EXFOR fission yield FY(A,Ei): 92-U-238(N,F)MASS,CHN,FY

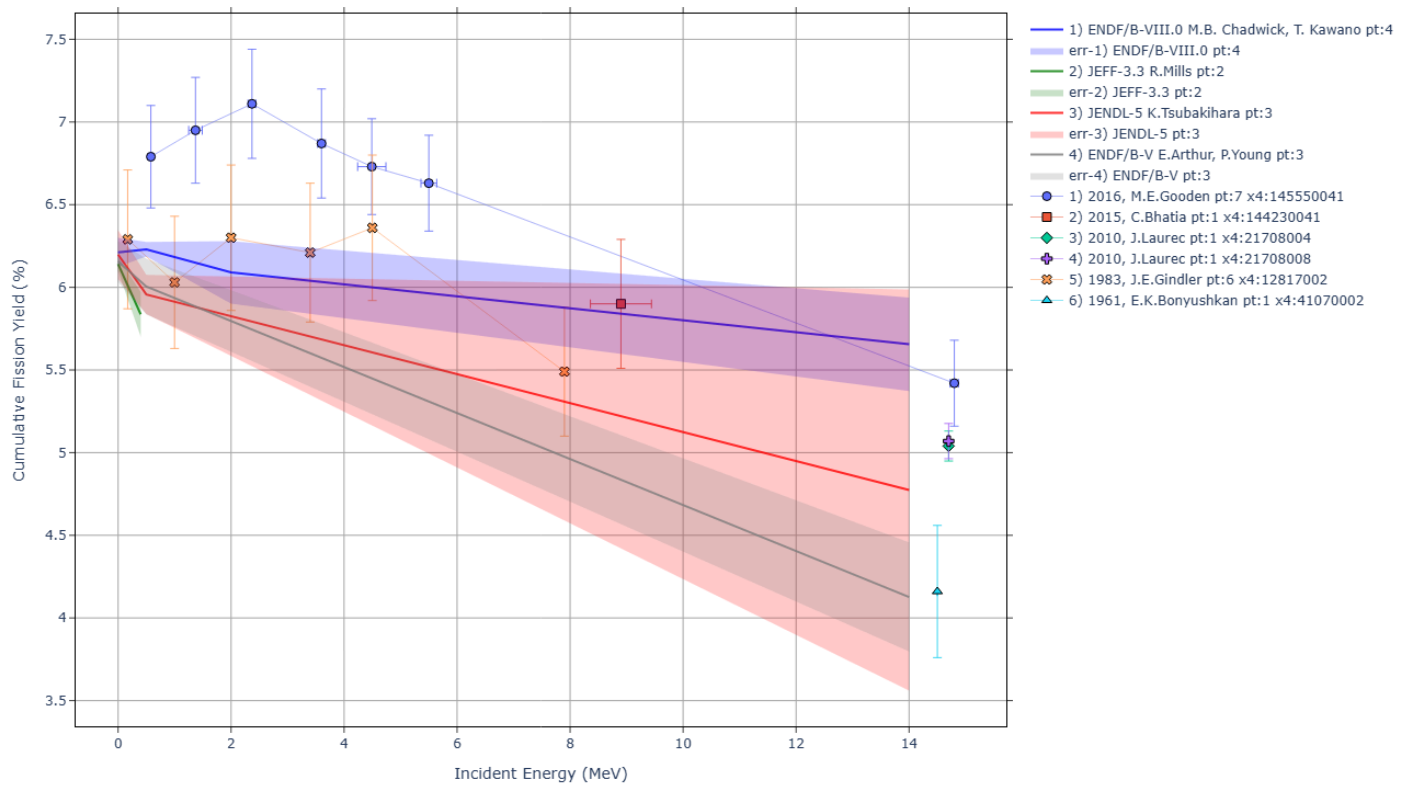
X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:08:11



part2-7-fye/ EXFOR/ENDF fission yeild (energy distribution)

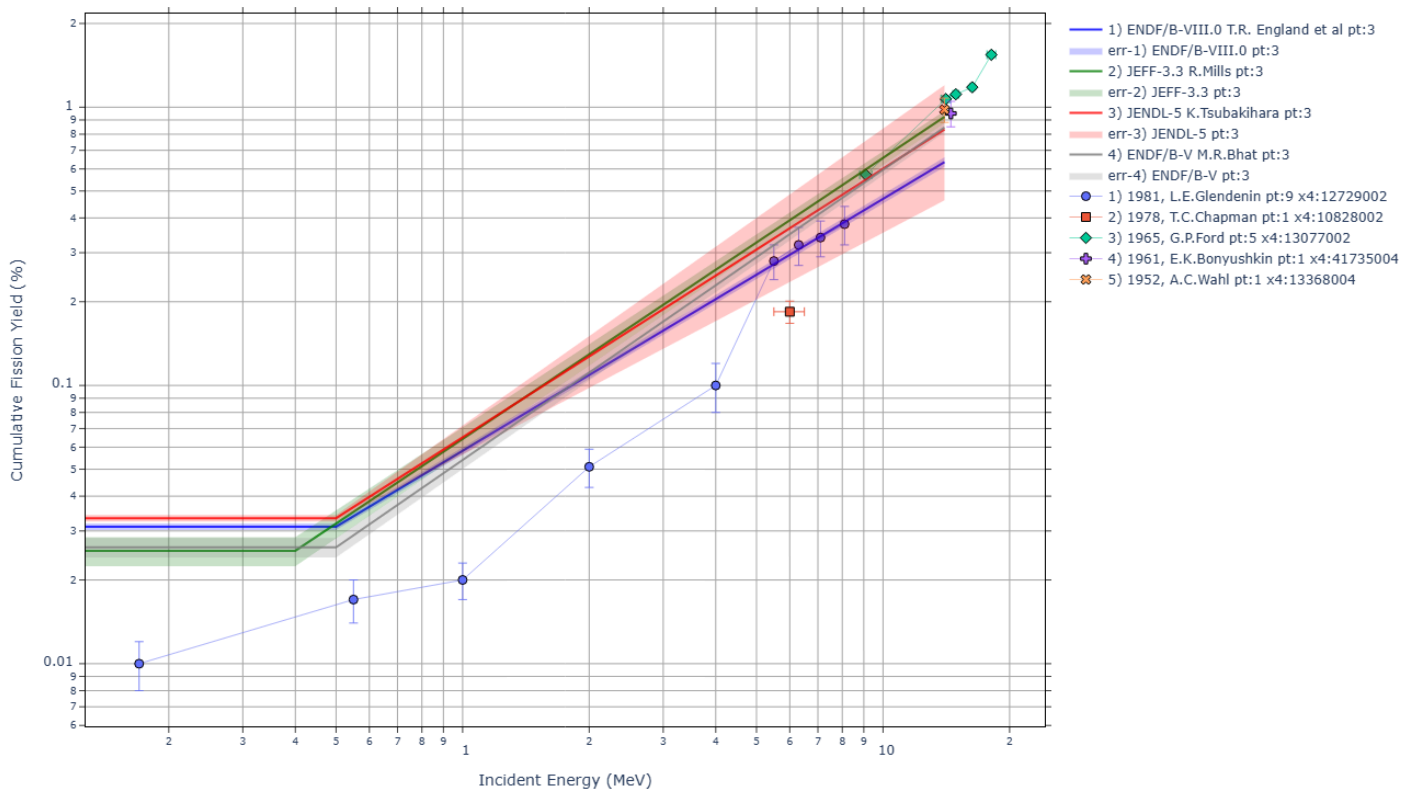
17) part2-7-fye/out00/Pu239-nf-Mo99-cumFY.html.png

EXFOR/ENDF cumulative fission yield FY(E): **94-PU-239(N,F)42-MO-99,CUM,FY**
 X4Pro, by V.Zerkin, IAEA, 2021-2024, ver.2024-09-02 //run:2026-08-21 12:08:20

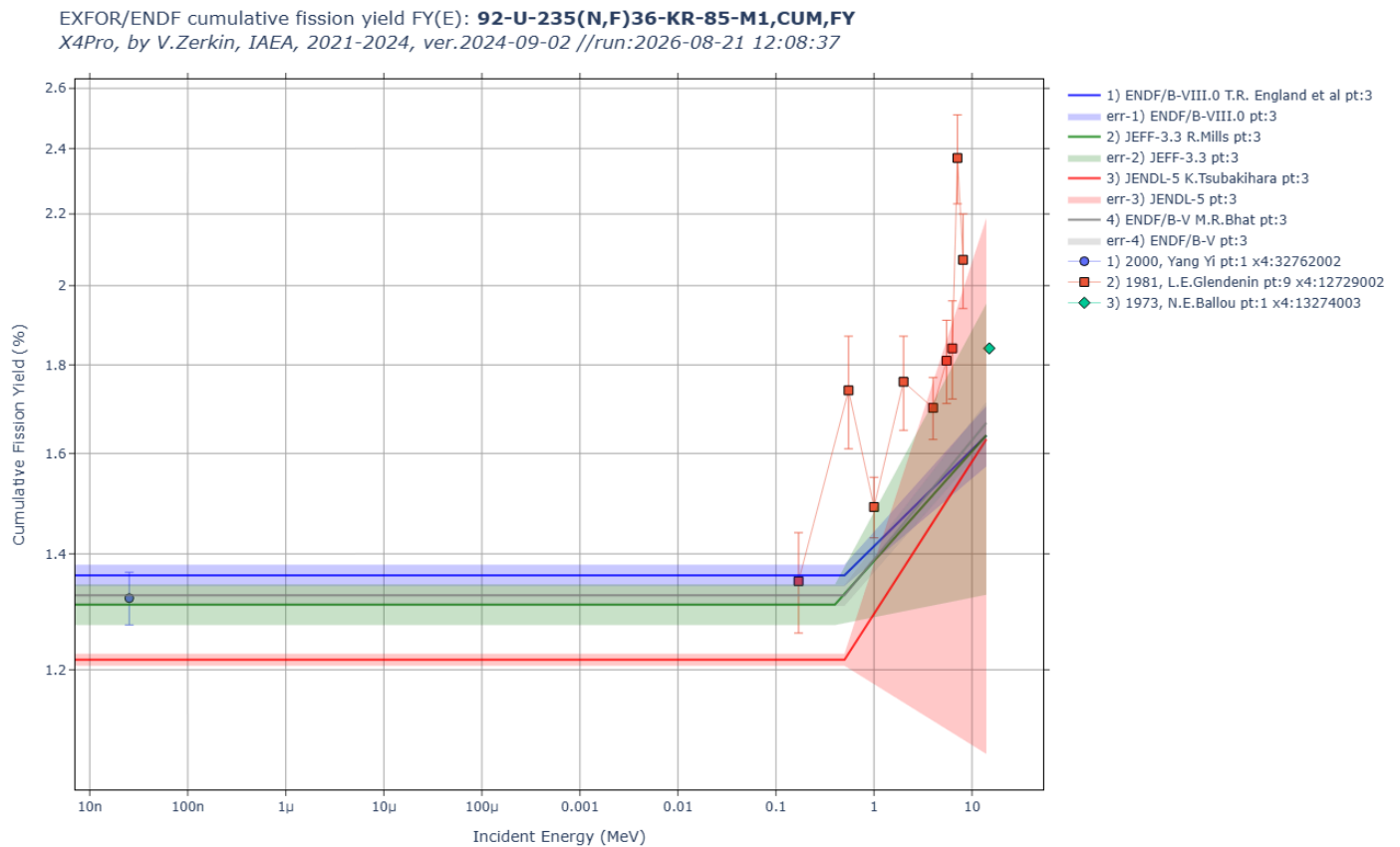


18) part2-7-fye/out00/u235-nf-cd115g-cumFY.html.png

EXFOR/ENDF cumulative fission yield FY(E): **92-U-235(N,F)48-CD-115-G,CUM,FY**
 X4Pro, by V.Zerkin, IAEA, 2021-2024, ver.2024-09-02 //run:2026-08-21 12:08:53

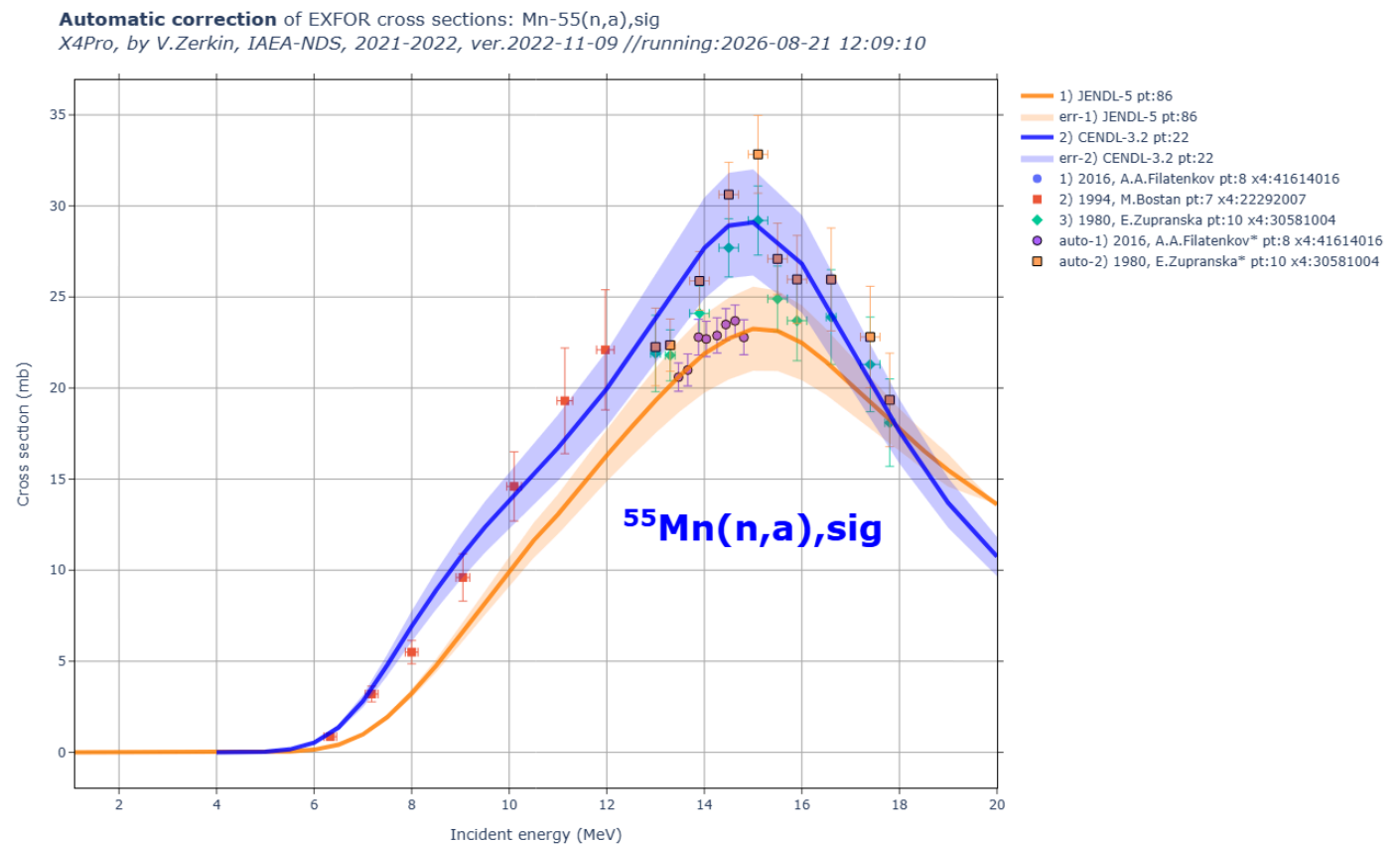


19) part2-7-fye/out00/u235-nf-kr85m-cumFY.html.png



part3-1-auto1/ Automatically renormalize EXFOR CS using old monitor CS and new standards

20) part3-1-auto1/out00/auto1.html.png

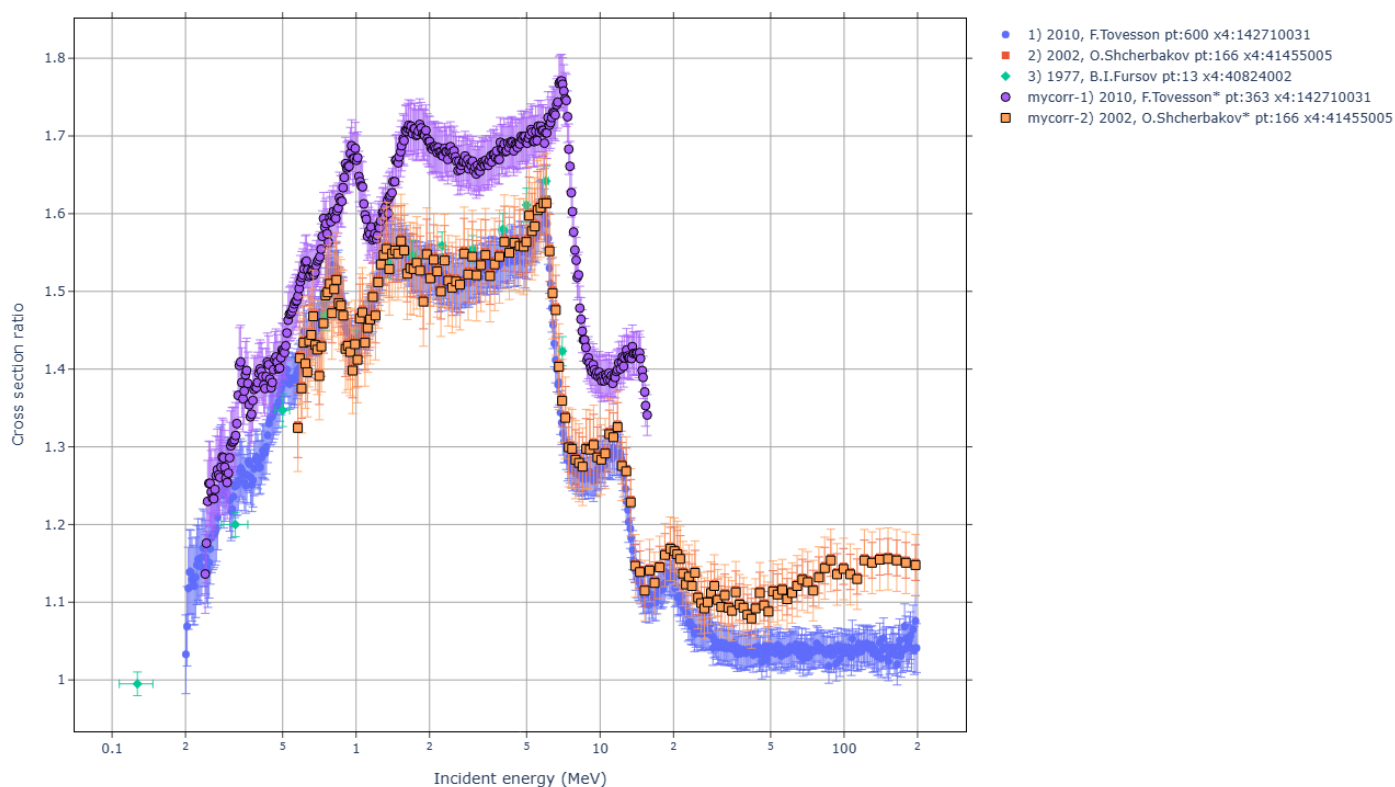


part3-2-user1/ *Correct data from EXFOR by user's program*

21) part3-2-user1/out00/user1.html.png

Local user's corrections of EXFOR cross sections ratios: Pu-239/U-235(n,f)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:09:21

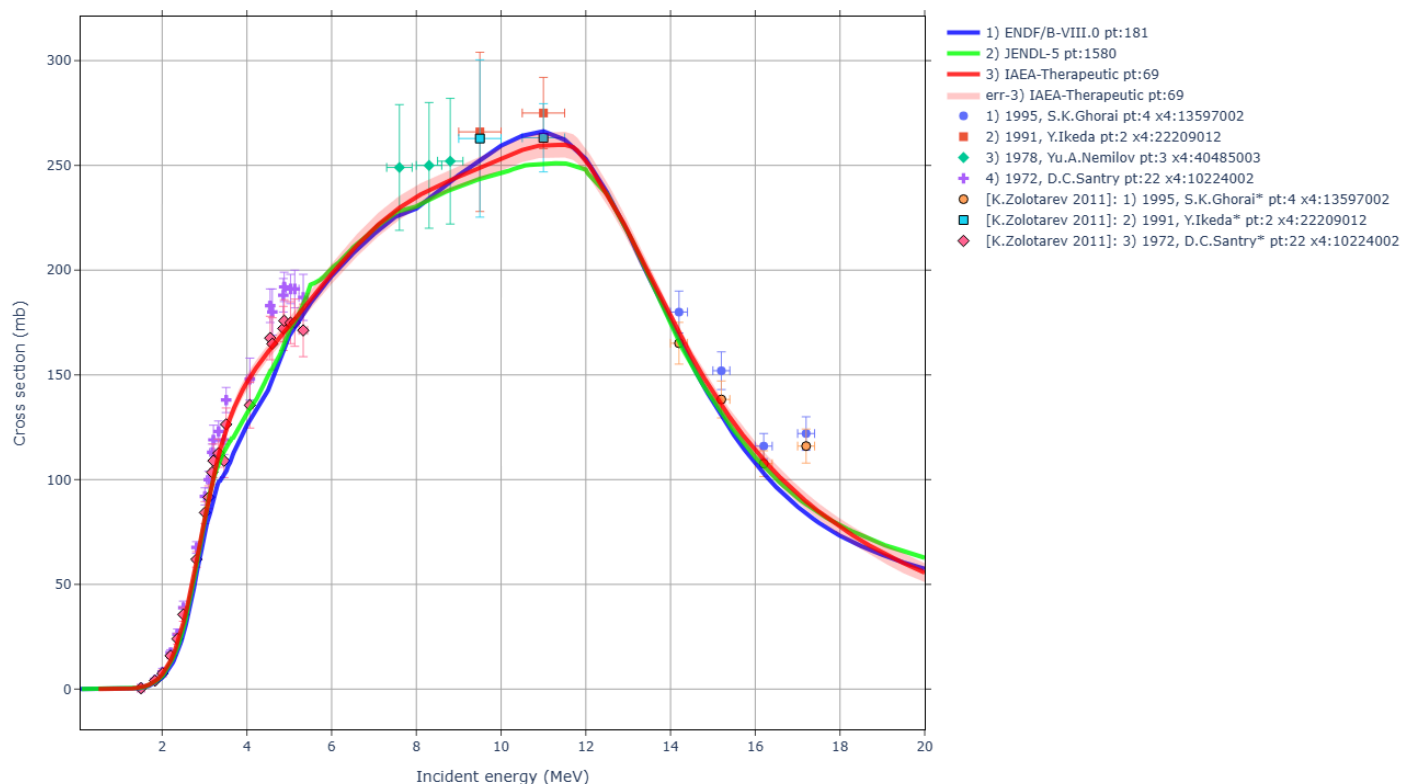


part3-3-expert1/ *Apply experts corrections stored in X4Pro database*

22) part3-3-expert1/out00/expert1.html.png

Apply experts corrections from database to EXFOR data: Zn-64(n,p),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:09:24

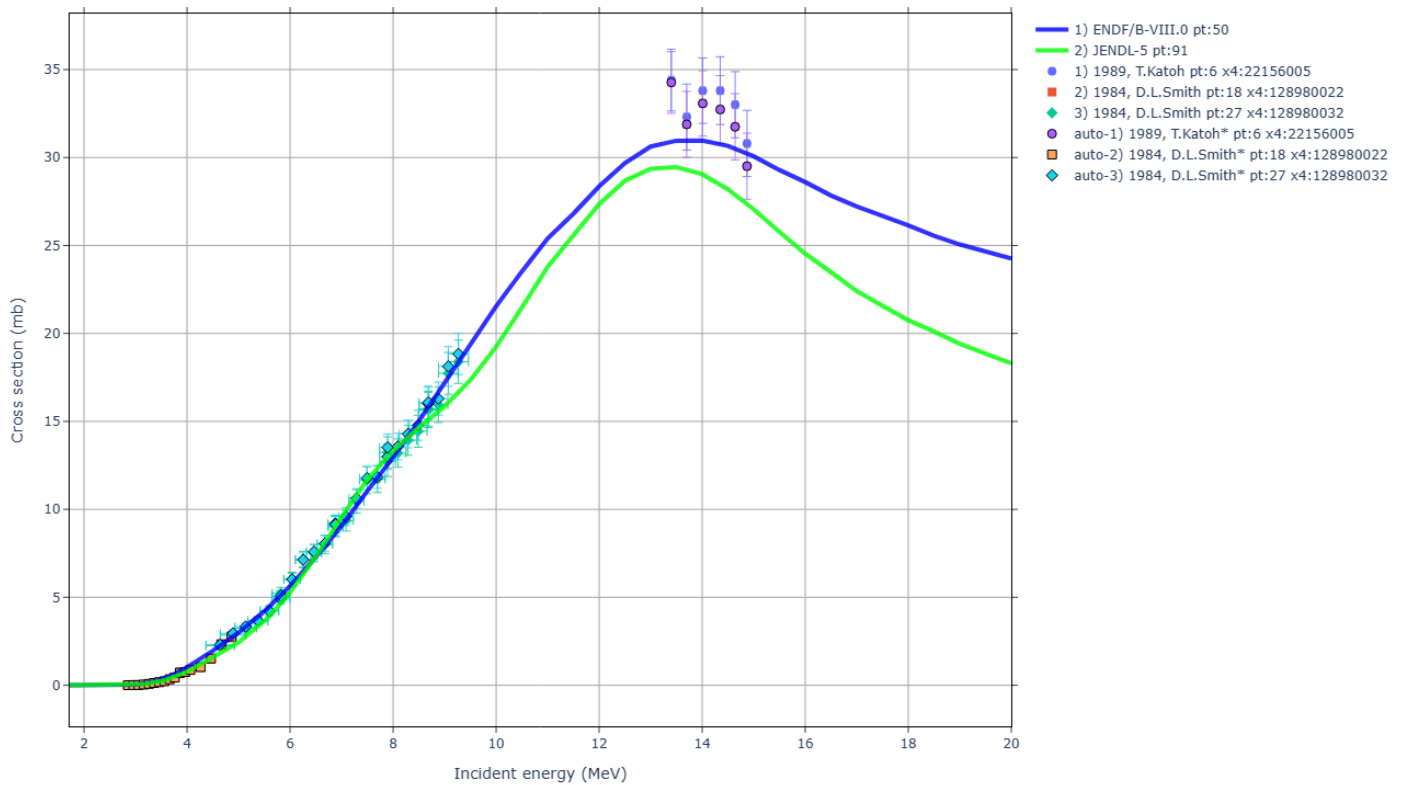


part3-4-auto2/ Automatically renormalize EXFOR CS using old monitor CS and new standards, old and new decay data

23) part3-4-auto2/out00/auto2.html.png

Automatic correction of EXFOR cross sections: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:09:38

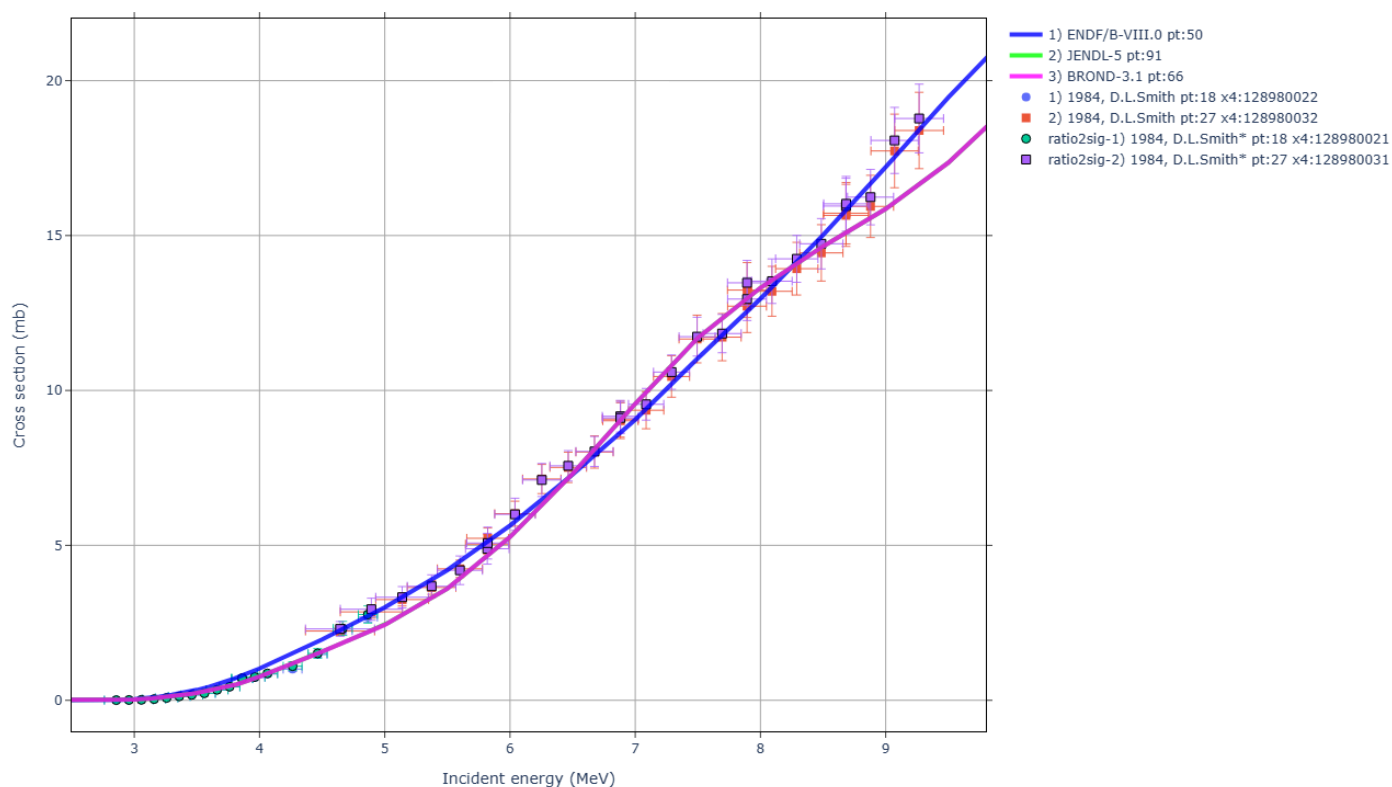


part3-5-ratio2sig/ Retrieve ratios, calculate CS and uncertainties using latest standards

24) part3-5-ratio2sig/out00/ratio2cs.html.png

Ratio to cross section of EXFOR data: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-08-21 12:09:49

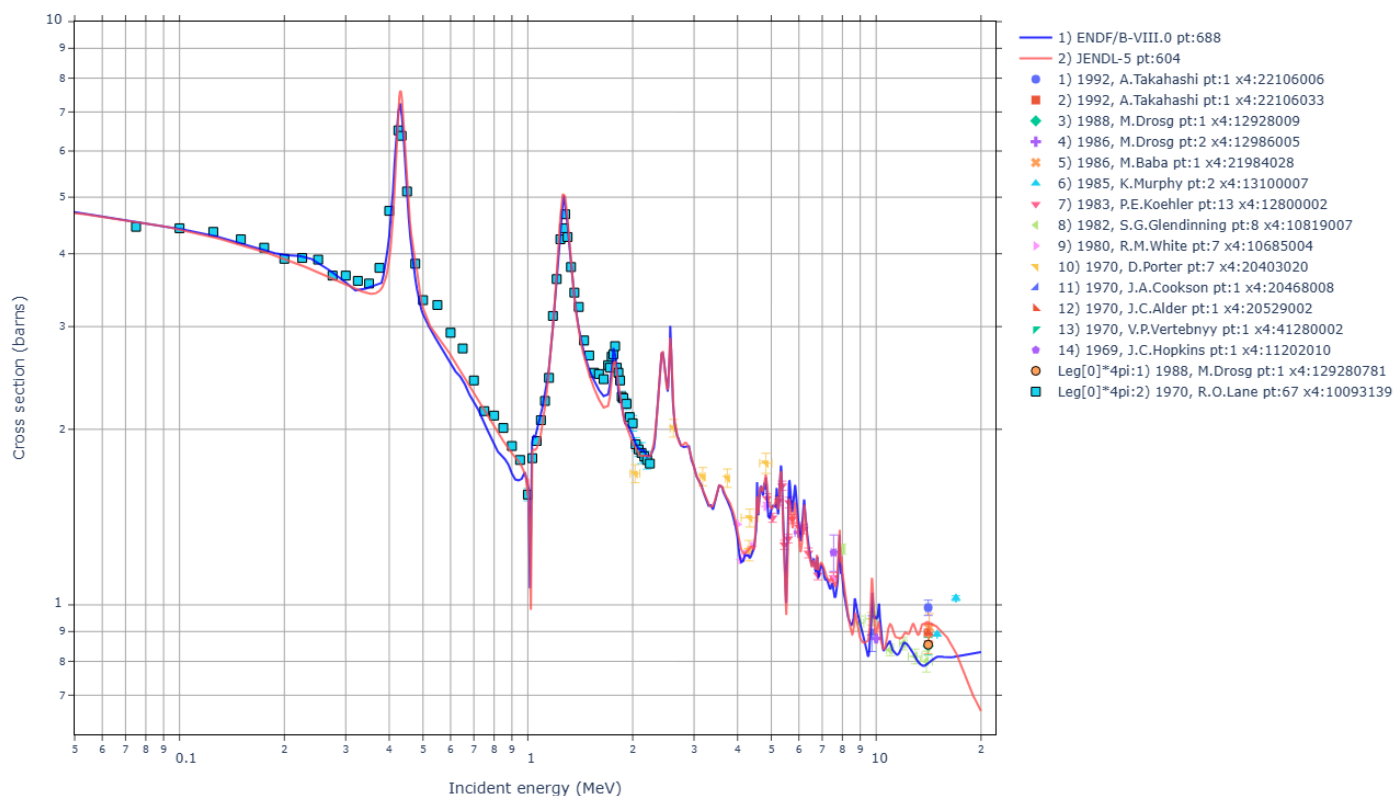


part3-6-daleg2sig/ Retrieve from EXFOR: Legendre coefficient L[0], calculate cross sections and errors

25) part3-6-daleg2sig/out00/B11nel.html.png

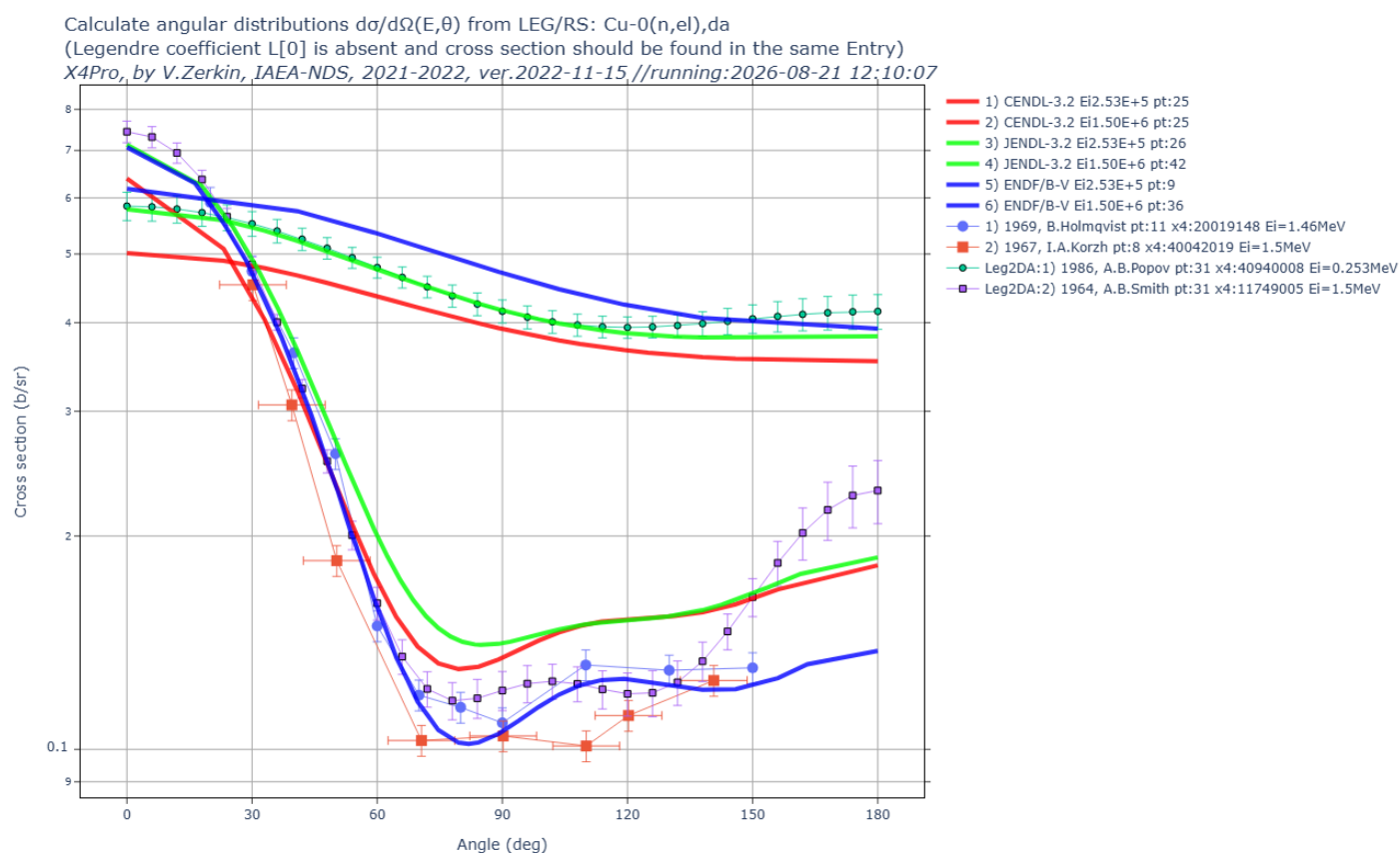
Cross sections $\sigma(E)$: from B-11(n,el),SIG and DA,,LEG:Legendre coeff.L[0]*4pi

X4Pro, by V.Zerkin, IAEA-NDS, 2021-12-07--2022-11-09 //running:2026-08-21 12:09:55



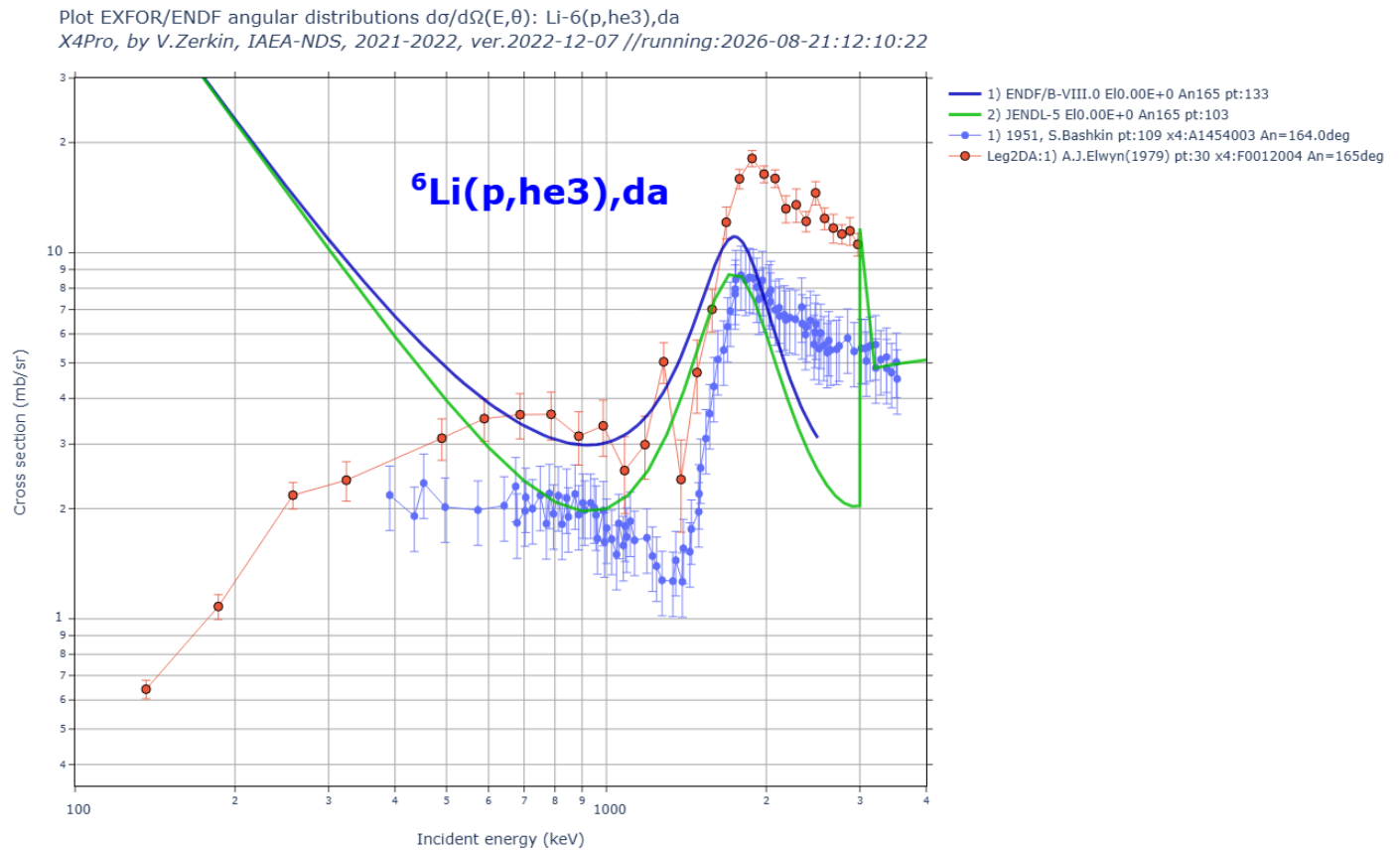
part3-7-legrs2da/ EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre coefficient $L[0]$; calc. angular distributions

26) part3-7-legrs2da/out00/Cu0nel_da.html.png



part3-8-leg2r33/ Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)

27) part3-8-leg2r33/out00/da1ei_leg2da.html.png



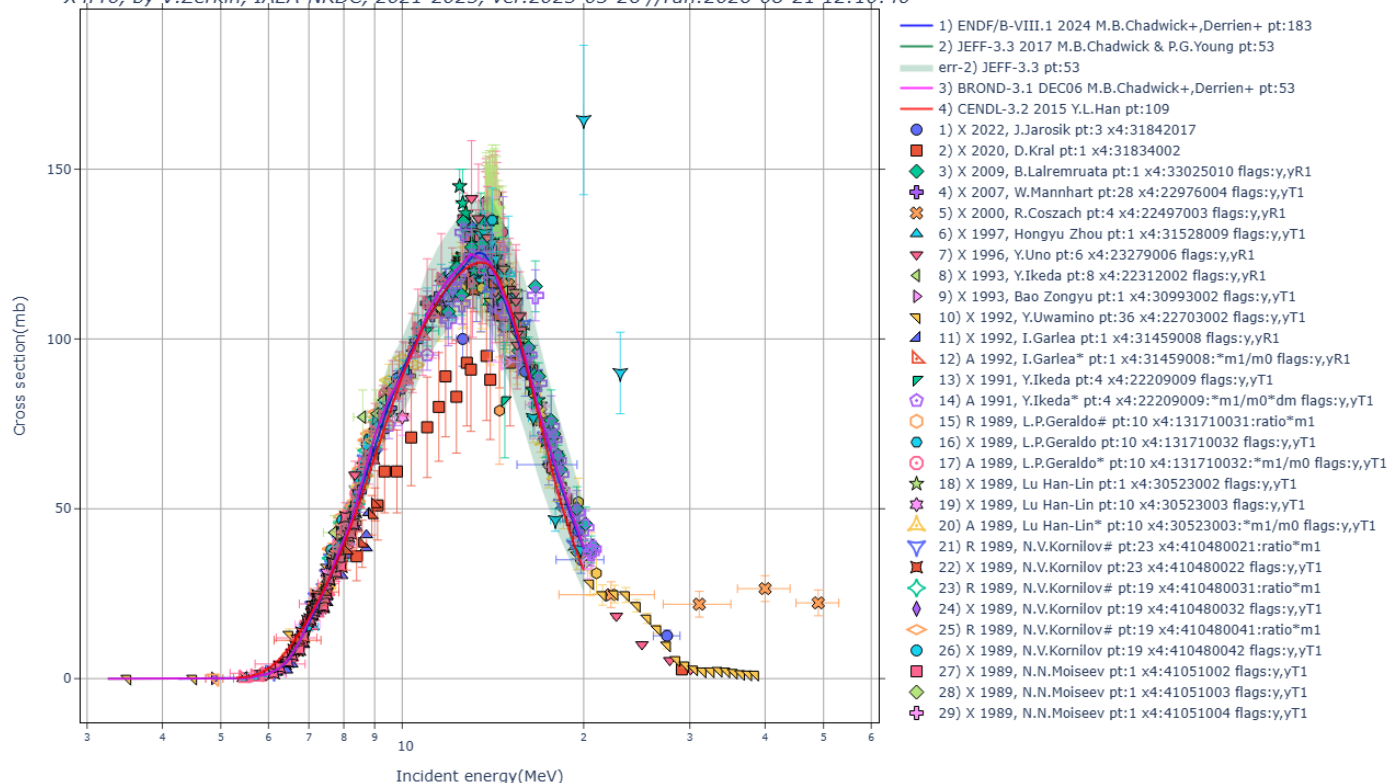
part6-1-eval2sig/ Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS

28) part6-1-eval2sig/out00/Al27na.html.png

Cross sections $\sigma(E)$: Al-27(n,a),sig datasets:116 original:85X auto-corrected:22A* ratio2cs:9R#

Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-21 12:10:40

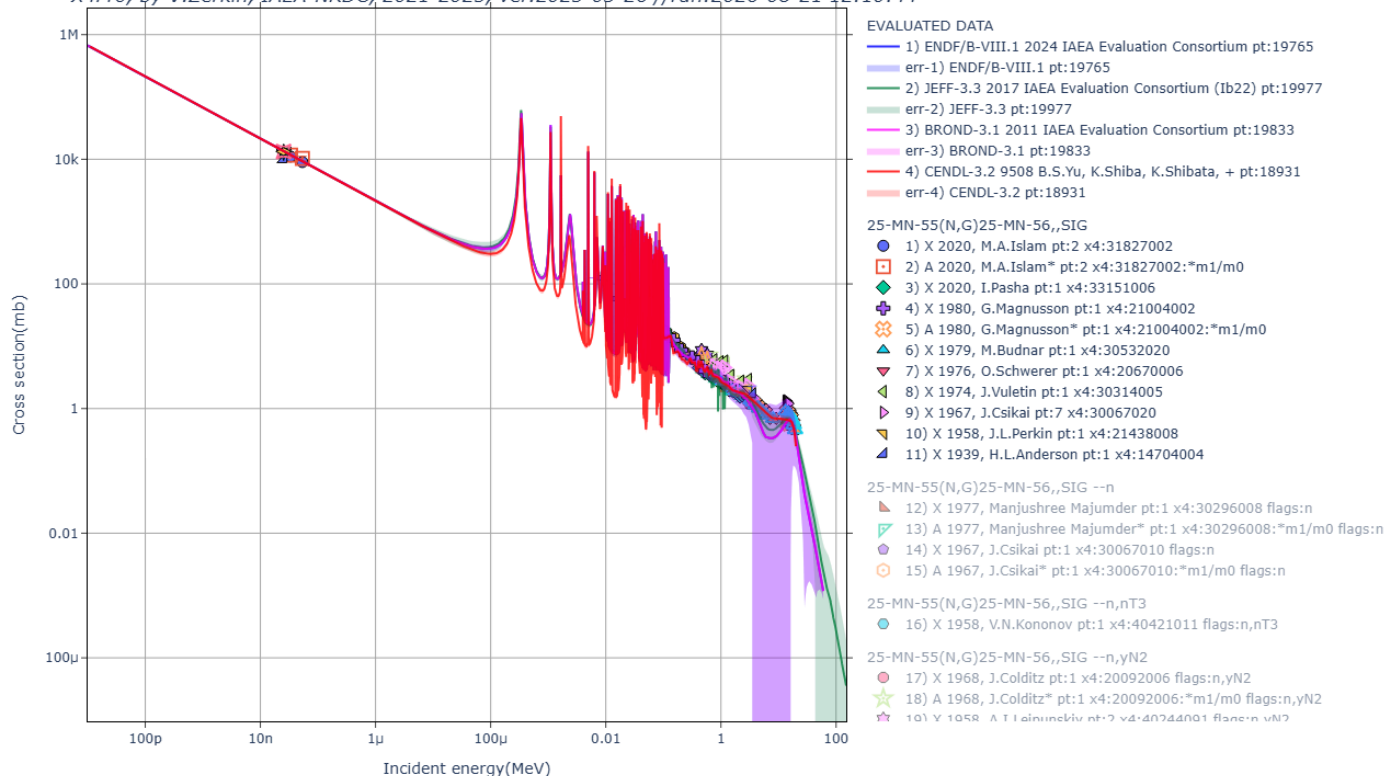


29) part6-1-eval2sig/out00/Mn55ng-n.html.png

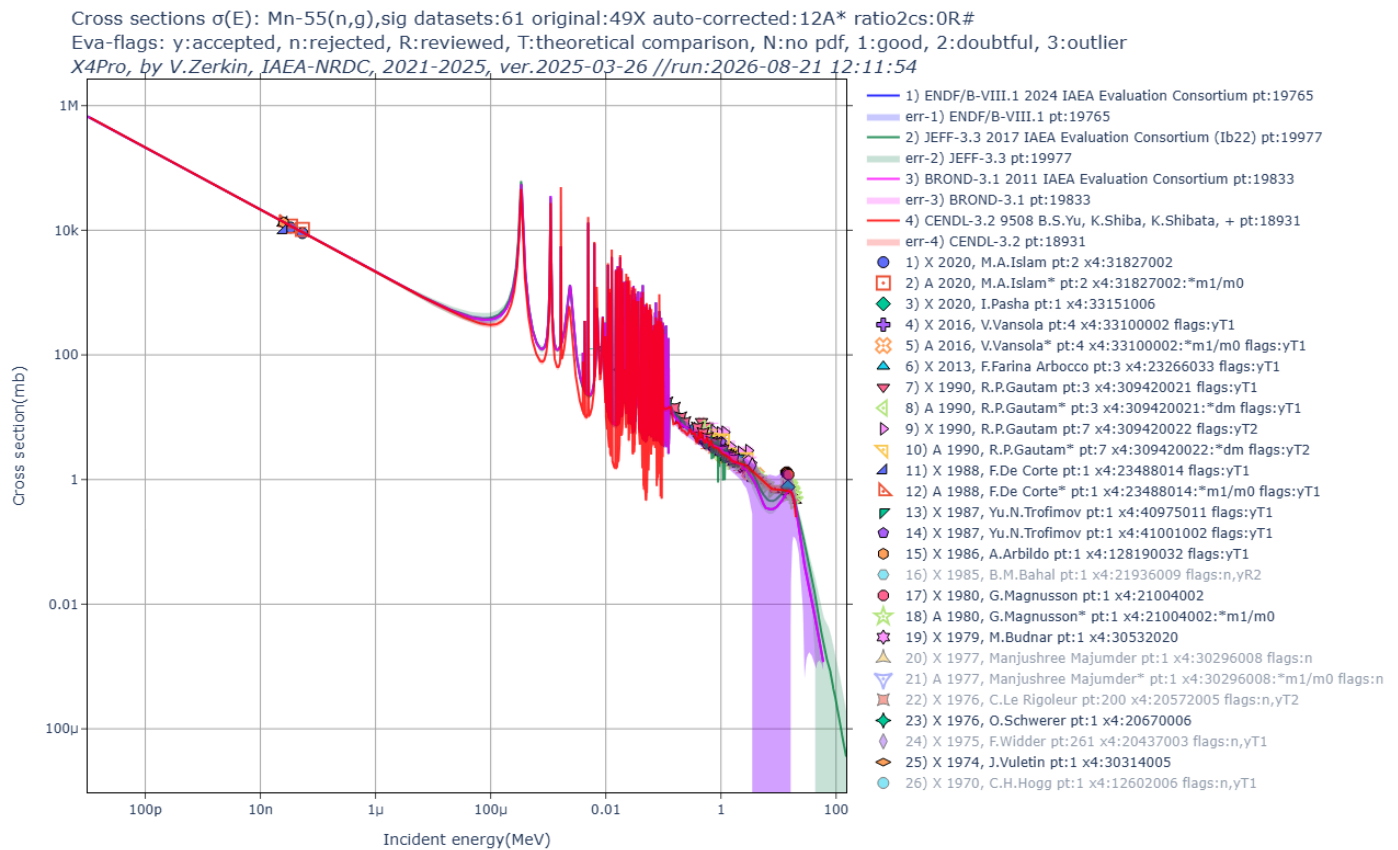
Cross sections $\sigma(E)$: Mn-55(n,g),sig datasets:61 original:49X auto-corrected:12A* ratio2cs:0R#

Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier

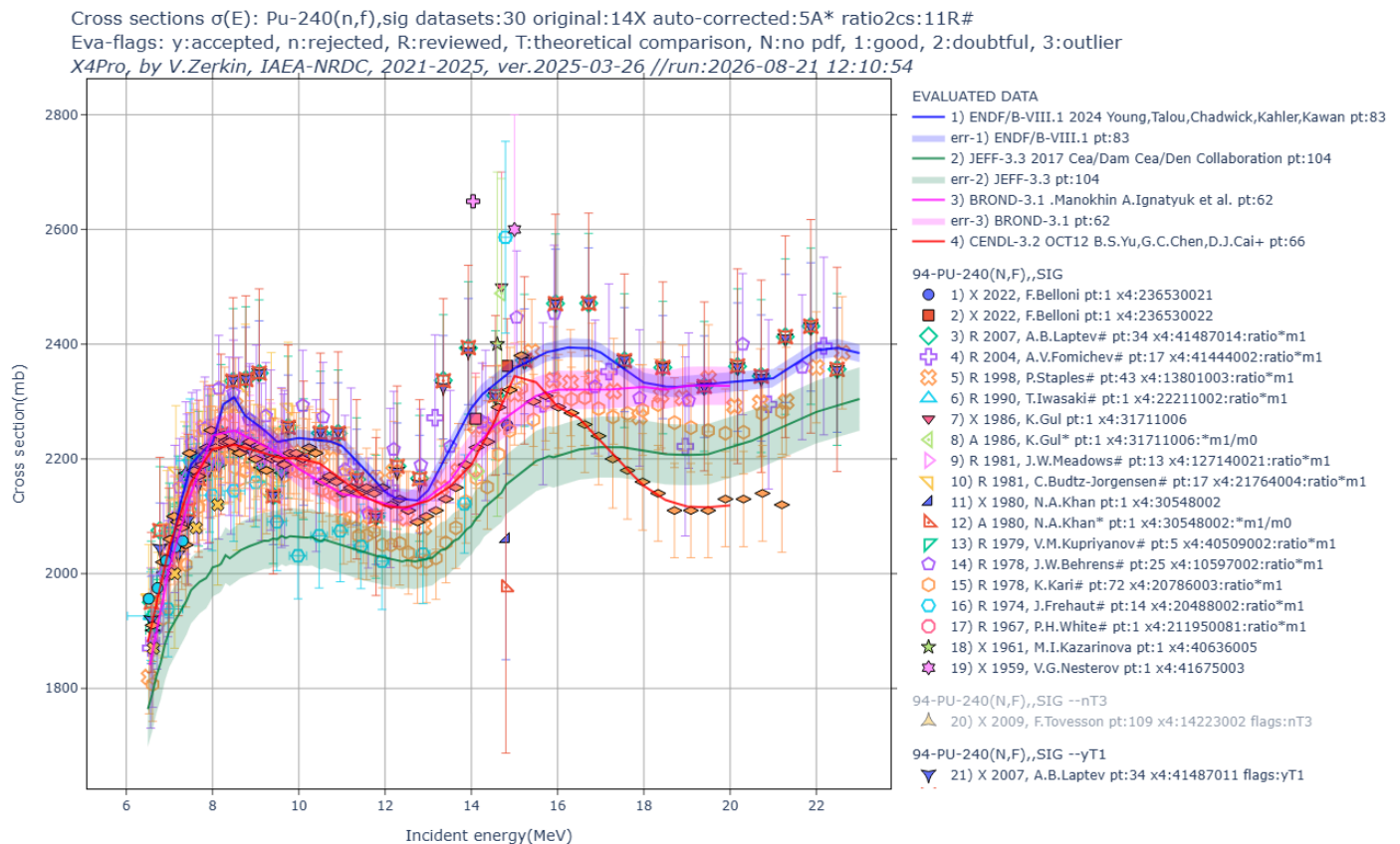
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-21 12:10:44



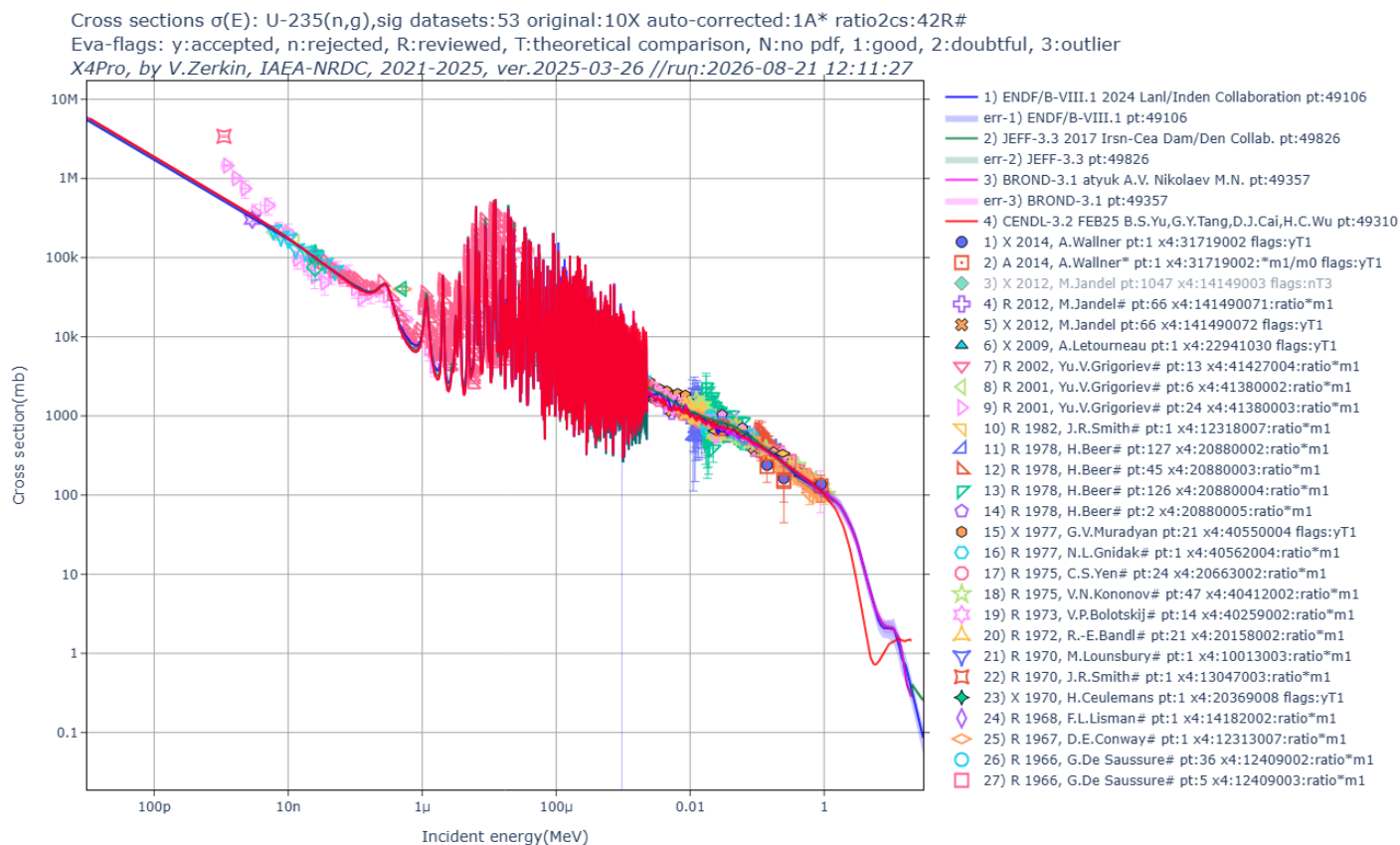
30) part6-1-eval2sig/out00/Mn55ng.html.png



31) part6-1-eval2sig/out00/Pu240nf-n.html.png



32) part6-1-eval2sig/out00/U235ng.html.png



33) part6-1-eval2sig/out00/Zn64np.html.png

